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Arc path formation unit and direct current relay including same

Abstract

An arc path formation unit and a direct current relay including same are disclosed. The arc path formation unit according to various embodiments of the present disclosure comprises a Halbach array and a magnet unit, which form a magnetic field in a space part in which fixed contacts are accommodated. The formed magnetic field forms an electromagnetic force together with the current applied to the direct current relay. The formed electromagnetic force can induce generated arcs here, the electromagnetic force formed in proximity to the respective fixed contacts is formed in the direction of moving away from the respective fixed contacts. Therefore, the generated arcs do not meet each other, and thus can be effectively extinguished and discharged.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a National Stage of International Application No. PCT/KR2021/006517 filed on May 25, 2021 claims priority to and the benefit of Korean Patent Application No. 10-2020-0079614, filed on Jun. 29, 2020, the disclosures of which are incorporated herein by reference in their entirety.

FIELD

(2) The present disclosure relates to an arc path formation unit and a direct current relay including the same, and more particularly, to an arc path formation unit having a structure capable of effectively inducing a generated arc toward the outside and a direct current relay including the same.

BACKGROUND

(3) A direct current relay is a device that transmits a mechanical driving signal or a current signal using the principle of an electromagnet. The direct current relay is also called a magnetic switch and is generally classified as an electrical circuit switching device.

(4) The direct current relay includes a fixed contact and a movable contact. The fixed contact is electrically connected to an external power supply and a load. The fixed contact and the movable contact may be brought into contact with or separated from each other.

(5) By the contact and separation between the fixed contact and the movable contact, a current flow through the direct current relay is allowed or blocked. Such a movement is made by a driving unit that applies a driving force to the movable contact.

(6) When the fixed contact and the movable contact are separated from each other, an arc is generated between the fixed contact and the movable contact. The arc is a flow of high-pressure and high-temperature current. Accordingly, the generated arc must be quickly discharged from the direct current relay through a predetermined path.

(7) An arc discharge path is formed by magnets provided in the direct current relay. The magnets form magnetic fields in a space in which the fixed contact and the movable contact are in contact with each other. The arc discharge path may be formed by the formed magnetic field and an electromagnetic force generated by a flow of current.

(8) Referring to FIG. **1**, a space in which fixed contacts **1100** and a movable contact **1200** provided in a direct current relay **1000** according to the related art are in contact with each other is illustrated. As described above, permanent magnets **1300** are provided in the space.

(9) The permanent magnets **1300** include a first permanent magnet **1310** disposed at an upper side and a second permanent magnet **1320** disposed at a lower side.

(10) The first permanent magnet **1310** is provided in plural, and each surface facing the second

permanent magnet **1320** is magnetized to a different polarity. A lower side of the first permanent magnet **1310** located on a left side of FIG. **1** is magnetized to an N pole, and a lower side of the first permanent magnet **1310** located on a right side of FIG. **1** is magnetized to an S pole.
(11) In addition, the second permanent magnet **1320** is also provided in plural, and each surface facing the first permanent magnet **1310** is magnetized to a different polarity. An upper side of the second permanent magnet **1320** located on the left side of FIG. **1** is magnetized to an S pole, and an upper side of the second permanent magnet **1320** located on the right side of FIG. **1** is magnetized to an N pole.
(12) FIG. **1**A illustrates a state in which current flows in through the left fixed contact **1100** and flows out through the right fixed contact **1100**. According to the Fleming's left-hand rule, an electromagnetic force is formed as indicated by hatched arrows.
(13) Specifically, in the case of the fixed contact **1100** located on the left side, the electromagnetic force is formed toward the outside. Accordingly, the arc generated at the corresponding location can be discharged to the outside.
(14) However, in the case of the fixed contact **1100** located on the right side, the electromagnetic force is formed to the inside, that is, toward a central portion of the movable contact **1200**. Accordingly, the arc generated at the corresponding location cannot be immediately discharged to the outside.
(15) In addition, FIG. **1**B illustrates a state in which current flows in through the right fixed contact **1100** and flows out through the left fixed contact **1100**. According to the Fleming's left-hand rule, an electromagnetic force is formed as indicated by hatched arrows.
(16) Specifically, in the case of the fixed contact **1100** located on the right side, the electromagnetic force is formed toward the outside. Accordingly, the arc generated at the corresponding location can be discharged to the outside.
(17) However, in the case of the fixed contact **1100** located on the left side, the electromagnetic force is formed to the inside, that is, toward the central portion of the movable contact **1200**. Accordingly, the arc generated at the corresponding location cannot be immediately discharged to the outside.
(18) Several members for driving the movable contact **1200** to be moved in a vertical direction are provided in a central portion of the direct current relay **1000**, that is, in a space between the fixed contacts **1100**. As an example, a shaft, a spring member inserted through the shaft, and the like are provided at the location.
(19) Accordingly, when the arc generated as illustrated in FIG. **1** is moved toward the central portion, and the arc moved to the central portion cannot be immediately moved to the outside, there is a risk that the several members provided at the location may be damaged by energy of the arc.
(20) In addition, as illustrated in FIG. **1**, a direction of the electromagnetic force formed inside the direct current relay **1000** according to the related art depends on a direction of current flowing through the fixed contacts **1100**. That is, the location of the electromagnetic force, which is formed in a direction toward the inside, among the electromagnetic forces generated in each fixed contact **1100** is different depending on the direction of the current.
(21) That is, a user must consider the direction of the current whenever using the direct current relay. This may cause inconvenience to the use of the direct current relay. In addition, regardless of the user's intention, a situation in which a direction of current applied to the direct current relay is changed due to an inexperienced operation or the like cannot be excluded.
(22) In this case, the members provided in the central portion of the direct current relay may be damaged by the generated arc. Accordingly, there is a concern of reducing the durable lifetime of the direct current relay and also generating safety accidents.
(23) Korean Registration Application No. 10-1696952 discloses a direct current relay. Specifically, a direct current relay having a structure capable of preventing movement of a movable contact by using a plurality of permanent magnets is disclosed.

(24) However, the direct current relay having the above structure can prevent the movement of the movable contact by using the plurality of permanent magnets, but there is a limitation in that any method for controlling a direction of an arc discharge path is not considered.
(25) Korean Registration Application No. 10-1216824 discloses a direct current relay. Specifically, a direct current relay having a structure capable of preventing arbitrary separation between a movable contact and a fixed contact using a damping magnet is disclosed.
(26) However, the direct current relay having the above structure merely proposes a method for maintaining a contact state between the movable contact and the fixed contact. That is, there is a limitation in that a method for forming a discharge path for an arc generated when the movable contact and the fixed contact are separated from each other is not introduced.
(27) (Patent Document 1) Korean Registration Application No. 10-1696952 (Jan. 16, 2017)
(28) (Patent Document 2) Korean Registration Application No. 10-1216824 (Dec. 28, 2012)
SUMMARY
(29) The present disclosure is directed to providing an arc path formation unit having a structure capable of solving the above-described problems and a direct current relay including the same.
(30) First, the present disclosure is directed to providing an arc path formation unit having a structure capable of quickly extinguishing and discharging an arc generated as flowing current is interrupted, and a direct current relay including the same.
(31) In addition, the present disclosure is directed to providing an arc path formation unit having a structure capable of increasing the magnitude of force for inducing a generated arc, and a direct current relay including the same.
(32) In addition, the present disclosure is directed to providing an arc path formation unit having a structure capable of preventing damage to a component for electric connection due to a generated arc, and a direct current relay including the same.
(33) In addition, the present disclosure is directed to providing an arc path formation unit having a structure capable of allowing arcs generated at a plurality of locations to propagate without meeting each other, and a direct current relay including the same.
(34) In addition, the present disclosure is directed to providing an arc path formation unit having a structure capable of achieving the above-described objects without an excessive design change, and a direct current relay including the same.
(35) In order to achieve those objects, one embodiment of the present disclosure provides an arc path formation unit including a magnet frame having a space part, in which a fixed contactor and a movable contactor are accommodated, formed therein, and a Halbach array located in the space part of the magnet frame and configured to form a magnetic field in the space part, wherein a length of the space part in one direction is formed to be greater than a length thereof in the other direction, the magnet frame includes a first surface and a second surface which extend in the one direction, are disposed to face each other, and are configured to surround a portion of the space part, and a third surface and a fourth surface which extend in the other direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, the fixed contactor includes a first fixed contactor located to be biased to one side in the one direction, and a second fixed contactor located to be biased to the other side in the one direction, and the Halbach array includes a first Halbach array located adjacent to any one surface of the first surface and the second surface, located to be biased to any one surface of the third surface and the fourth surface, and disposed to overlap any one contactor of the first fixed contactor and the second fixed contactor in the other direction.
(36) In addition, the Halbach array of the arc path formation unit may include a second Halbach array located adjacent to the other surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to face the first Halbach array with the space part therebetween, wherein the first Halbach array and the

second Halbach array may be disposed to overlap the any one contactor of the first fixed contactor and the second fixed contactor in the other direction.
(37) In addition, surfaces of the first Halbach array and the second Halbach array of the arc path formation unit facing each other may be magnetized to the same polarity.
(38) In addition, the Halbach array of the arc path formation unit may include a third Halbach array located adjacent to the other surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction.
(39) In addition, surfaces of the first Halbach array and the second Halbach array of the arc path formation unit facing each other may be magnetized to the same polarity, and among surfaces of the third Halbach array, a surface facing the space part may be magnetized to a polarity the same as the polarity.
(40) In addition, the arc path formation unit may include a magnet unit provided separately from the Halbach array, located in the space part of the magnet frame to form a magnetic field in the space part, located adjacent to the any one surface of the third surface and the fourth surface, and disposed to overlap the fixed contactor and the third Halbach array in the one direction.
(41) In addition, surfaces of the first Halbach array and the second Halbach array of the arc path formation unit facing each other may be magnetized to the same polarity, among surfaces of the third Halbach array, a surface facing the space part may be magnetized to a polarity the same as the polarity, and among surfaces of the magnet unit, a surface facing the space part may be magnetized to a polarity different from the polarity.
(42) In addition, the arc path formation unit may include a magnet unit provided separately from the Halbach array, located in the space part of the magnet frame to form a magnetic field in the space part, located adjacent to the other surface of the third surface and the fourth surface, and disposed to overlap the fixed contactor in the one direction.
(43) In addition, surfaces of the first Halbach array and the second Halbach array of the arc path formation unit facing each other may be magnetized to the same polarity, and among surfaces of the magnet unit, a surface facing the space part may be magnetized to a polarity the same as the polarity.
(44) In addition, the arc path formation unit may include a magnet unit provided separately from the Halbach array and located in the space part of the magnet frame to form a magnetic field in the space part, wherein the magnet unit may include a first magnet unit located adjacent to the other surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to face the first Halbach array with the space part therebetween.
(45) In addition, surfaces of the first Halbach array and the first magnet unit of the arc path formation unit facing each other may be magnetized to the same polarity.
(46) In addition, the magnet unit of the arc path formation unit may include a second magnet unit located adjacent to the other surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction.
(47) In addition, surfaces of the first Halbach array and the first magnet unit of the arc path formation unit facing each other may be magnetized to the same polarity, and among surfaces of the second magnet unit, a surface facing the space part may be magnetized to a polarity the same as the polarity.
(48) In addition, the Halbach array of the arc path formation unit may include a second Halbach array located adjacent to the other surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction, and the magnet unit may include a second magnet unit located adjacent to the any one surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction.
(49) In addition, surfaces of the first Halbach array and the first magnet unit of the arc path formation unit facing each other may be magnetized to the same polarity, among surfaces of the

second Halbach array, a surface facing the space part may be magnetized to a polarity the same as the polarity, and among surfaces of the second magnet unit, a surface facing the space part may be magnetized to a polarity different from the polarity.

(50) In addition, another embodiment of the present disclosure provides an arc path formation unit including a magnet frame having a space part, in which a fixed contactor and a movable contactor are accommodated, formed therein, and a Halbach array and a magnet unit, which are located in the space part of the magnet frame and configured to form a magnetic field in the space part, the magnet unit being provided separately from the Halbach array, wherein a length of the space part in one direction is formed to be greater than a length thereof in the other direction, the magnet frame includes a first surface and a second surface which extend in the one direction, are disposed to face each other, and are configured to surround a portion of the space part, and a third surface and a fourth surface which extend in the other direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, the fixed contactor includes a first fixed contactor located to be biased to one side in the one direction, and a second fixed contactor located to be biased to the other side in the one direction, the Halbach array includes a first Halbach array located adjacent to any one surface of the first surface and the second surface, located to be biased to any one surface of the third surface and the fourth surface, and disposed to overlap any one contactor of the first fixed contactor and the second fixed contactor in the other direction, and the magnet unit includes a first magnet unit located adjacent to the any one surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to overlap the first Halbach array and the any one contactor of the first fixed contactor and the second fixed contactor in the other direction.

(51) In addition, the Halbach array of the arc path formation unit may include a second Halbach array located adjacent to the other surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction, and the magnet unit may include a second magnet unit located adjacent to the any one surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction.

(52) In addition, surfaces of the first Halbach array and the first magnet unit of the arc path formation unit facing each other may be magnetized to the same polarity, among surfaces of the second Halbach array, a surface facing the space part may be magnetized to a polarity the same as the polarity, and among surfaces of the second magnet unit, a surface facing the space part may be magnetized to a polarity different from the polarity.

(53) In addition, still another embodiment of the present disclosure provides a direct current relay including a plurality of fixed contactors located to be spaced apart from each other in one direction, a movable contactor configured to be brought into contact with or separated from the fixed contactors, a magnet frame having a space part, in which the fixed contactors and the movable contactor are accommodated, formed therein, and a Halbach array located in the space part of the magnet frame and configured to form a magnetic field in the space part, wherein a length of the space part in one direction is formed to be greater than a length thereof in the other direction, the magnet frame includes a first surface and a second surface which extend in the one direction, are disposed to face each other, and are configured to surround a portion of the space part, and a third surface and a fourth surface which extend in the other direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, the fixed contactor includes a first fixed contactor located to be biased to one side in the one direction, and a second fixed contactor located to be biased to the other side in the one direction, and the Halbach array includes a first Halbach array located adjacent to any one surface of the first surface and the second surface, located to be biased to any one surface of the third surface and the fourth surface, and disposed to overlap any one contactor of the first fixed contactor and the second fixed contactor in the other direction.

(54) In addition, the Halbach array of the direct current relay may include a second Halbach array located adjacent to the other surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to face the first Halbach array with the space part therebetween, wherein the first Halbach array and the second Halbach array may be disposed to overlap the any one contactor of the first fixed contactor and the second fixed contactor in the other direction, and surfaces of the first Halbach array and the second Halbach array facing each other may be magnetized to the same polarity.
(55) In addition, the direct current relay may include a magnet unit provided separately from the Halbach array, and located in the space part of the magnet frame to form a magnetic field in the space part, wherein the magnet unit may include a first magnet unit located adjacent to the other surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to face the first Halbach array with the space part therebetween, and surfaces of the first Halbach array and the first magnet unit facing each other may be magnetized to the same polarity.
(56) In addition, yet another embodiment of the present disclosure provides a direct current relay including a plurality of fixed contactors located to be spaced apart from each other in one direction, a movable contactor configured to be brought into contact with or separated from the fixed contactors, a magnet frame having a space part, in which the fixed contactors and the movable contactor are accommodated, formed therein, and a Halbach array and a magnet unit, which are located in the space part of the magnet frame and configured to form a magnetic field in the space part, the magnet unit being provided separately from the Halbach array, wherein a length of the space part in one direction is formed to be greater than a length thereof in the other direction, the magnet frame includes a first surface and a second surface which extend in the one direction, are disposed to face each other, and are configured to surround a portion of the space part, and a third surface and a fourth surface which extend in the other direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, the fixed contactor includes a first fixed contactor located to be biased to one side in the one direction, and a second fixed contactor located to be biased to the other side in the one direction, the Halbach array includes a first Halbach array located adjacent to any one surface of the first surface and the second surface, located to be biased to any one surface of the third surface and the fourth surface, and disposed to overlap any one contactor of the first fixed contactor and the second fixed contactor in the other direction, and the magnet unit includes a first magnet unit located adjacent to the any one surface of the first surface and the second surface, located to be biased to the any one surface of the third surface and the fourth surface, and disposed to overlap the first Halbach array and the any one contactor of the first fixed contactor and the second fixed contactor in the other direction.
(57) In addition, the Halbach array of the direct current relay may include a second Halbach array located adjacent to the other surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction, and the magnet unit may include a second magnet unit located adjacent to the any one surface of the third surface and the fourth surface and disposed to overlap the fixed contactor in the one direction, wherein surfaces of the first Halbach array and the first magnet unit facing each other may be magnetized to the same polarity, among surfaces of the second Halbach array, a surface facing the space part may be magnetized to a polarity the same as the polarity, and among surfaces of the second magnet unit, a surface facing the space part may be magnetized to a polarity different from the polarity.
(58) According to embodiments of the present disclosure, the following effects can be achieved.
(59) First, an arc path formation unit includes a Halbach array and a magnet unit. Each of the Halbach array and the magnet unit forms a magnetic field inside the arc path formation unit. The formed magnetic field forms an electromagnetic force together with current flowing through a fixed contactor and a movable contactor accommodated in the arc path formation unit.

(60) At this point, a generated arc is formed in a direction away from each fixed contactor. An arc generated as the fixed contactor and the movable contactor are separated from each other can be induced by the electromagnetic force.
(61) Accordingly, the generated arc can be quickly extinguished and discharged to the outside of the arc path formation unit and the direct current relay.
(62) In addition, the arc path formation unit includes a Halbach array. The Halbach array includes a plurality of magnetic materials disposed in parallel to each other in one direction. Each of the plurality of magnetic materials can enhance the strength of a magnetic field on any one side of both sides thereof in the other direction different from the one direction.
(63) At this point, the Halbach array is disposed such that the any one side, that is, the side in the direction in which the strength of the magnetic field is enhanced, faces a space part of the arc path formation unit. That is, due to the Halbach array, the strength of the magnetic field formed in the space part can be enhanced.
(64) Accordingly, the strength of the electromagnetic force, which depends on the strength of the magnetic field, can also be enhanced. As a result, the strength of the electromagnetic force inducing the generated arc can be enhanced so that the generated arc can be effectively extinguished and discharged.
(65) In addition, directions of the magnetic fields formed by the Halbach array and the magnet unit and a direction of the electromagnetic force formed by the current flowing through the fixed contactor and the movable contactor are formed to be away from a central part.
(66) Furthermore, as described above, since the strength of each of the magnetic field and the electromagnetic force is enhanced by the Halbach array and the magnet unit, the generated arc can be extinguished and moved quickly in a direction away from the central part.
(67) Accordingly, it is possible to prevent damage to various components that are provided in the vicinity of the central part for the operation of the direct current relay.
(68) In addition, in various embodiments, a plurality of fixed contactors can be provided. The Halbach array or the magnet units provided in the arc path formation unit forms magnetic fields in different directions in the vicinity of each fixed contactor. Thus, paths of the arc generated in the vicinity of each fixed contactor proceed in different directions.
(69) Accordingly, the arcs generated in the vicinity of each fixed contactor do not meet each other. Thus, a malfunction or a safety accident that may occur due to a collision of arcs generated at different locations can be prevented.
(70) In addition, in order to achieve the above-described objects and effects, the arc path formation unit includes a Halbach array and a magnet unit provided in a space part. Each of the Halbach array and the magnet unit is located on an inner side of each surface of a magnet frame surrounding the space part. That is, a separate design change for arranging the Halbach array and the magnet unit outside the space part is not required.
(71) Accordingly, without an excessive design change, the arc path formation unit according to various embodiments of the present disclosure can be provided in the direct current relay. Accordingly, time and costs for applying the arc path formation unit according to various embodiments of the present disclosure can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a conceptual view illustrating a direct current relay according to the related art.
(2) FIG. **2** is a perspective view illustrating a direct current relay according to an embodiment of the present disclosure.
(3) FIG. **3** is a cross-sectional view illustrating a configuration of the direct current relay of FIG. **2**.

(4) FIG. **4** is an opened perspective view illustrating an arc path formation unit provided in the direct current relay of FIG. **2**.
(5) FIGS. **5** and **6** are conceptual views illustrating the arc path formation unit according to one embodiment of the present disclosure.
(6) FIGS. **7** and **8** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **5** and **6**.
(7) FIGS. **9** and **10** are conceptual views illustrating an arc path formation unit according to another embodiment of the present disclosure.
(8) FIGS. **11** and **12** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **9** and **10**.
(9) FIGS. **13** to **16** are conceptual views illustrating an arc path formation unit according to still another embodiment of the present disclosure.
(10) FIGS. **17** to **20** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **13** to **16**.
(11) FIGS. **21** and **22** are conceptual views illustrating an arc path formation unit according to yet another embodiment of the present disclosure.
(12) FIGS. **23** and **24** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **21** and **22**.
(13) FIGS. **25** and **26** are conceptual views illustrating an arc path formation unit according to yet another embodiment of the present disclosure.
(14) FIGS. **27** and **28** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **25** and **26**.
(15) FIGS. **29** to **32** are conceptual views illustrating an arc path formation unit according to yet another embodiment of the present disclosure.
(16) FIGS. **33** to **36** are conceptual views illustrating magnetic fields and arc paths formed by the arc path formation unit according to the embodiment of FIGS. **29** to **32**.

DETAILED DESCRIPTION

(17) Hereinafter, a direct current relay **1** and arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to embodiments of the present disclosure will be described with reference to the accompanying drawings.
(18) In the following description, descriptions of some components may be omitted to clarify the features of the present disclosure.

1. Definition of Terms

(19) It will be understood that when a component is referred to as being "connected" or "coupled" to another component, it can be directly connected or coupled to the another component or intervening components may be present.
(20) In contrast, when a component is referred to as being "directly connected" or "directly coupled" to another component, there are no intervening components present.
(21) A singular representation used herein includes a plural representation unless it represents a definitely different meaning from the context.
(22) The term "magnetize" used in the following description means a phenomenon in which an object exhibits magnetism in a magnetic field.
(23) The term "polarities" used in the following description means different properties belonging to an anode and a cathode. In one embodiment, the polarities may be classified into an N pole or an S pole.
(24) The term "electric connection" used in the following description means a state in which two or more members are electrically connected.
(25) The term "arc path A.P" used in the following description means a path through which a generated arc is moved or extinguished.
(26) The symbol "⊙" shown in the following drawings means that current flows in a direction from

a movable contactor **43** toward a fixed contactor **22** (i.e., in an upward direction), that is, in a direction in which the current flows from the ground.
(27) The symbol ".Math." shown in the following drawings means that current flows in a direction from the fixed contactor **22** toward the movable contactor **43** (i.e., in a downward direction), that is, a direction in which the current flows into the ground.
(28) The term "Halbach array" used in the following description means an assembly of a plurality of magnetic materials that are disposed in parallel to form columns or rows.
(29) The plurality of magnetic materials constituting the Halbach array may be disposed according to a predetermined rule. A magnetic field may be formed by the magnetic material itself, or magnetic fields may also be formed by between the plurality of magnetic materials.
(30) The Halbach array includes two relatively long surfaces and two relatively short surfaces. Among the magnetic fields formed by the magnetic materials constituting the Halbach array, the magnetic field on an outer side of any one surface of the two long surfaces may be formed with a higher strength.
(31) In the following description, description will be made on the assumption that, among the magnetic fields formed by the Halbach array, the magnetic field in a direction toward a space part **115**, **215**, **315**, **415**, **515**, or **615** is formed with a higher strength.
(32) The term "magnet unit" used in the following description means any type of object that is formed of a magnetic material and capable of forming a magnetic field. In one embodiment, the magnet unit may be provided as a permanent magnet, an electromagnet, or the like. It will be understood that the magnet unit is different from the magnetic material forming the Halbach array, that is, a magnetic material provided separately from the Halbach array.
(33) The magnet unit may form a magnetic field by itself or together with another magnetic material.
(34) The magnet unit may extend in one direction. Both end portions of the magnet unit in the one direction may be magnetized to different polarities (i.e., the magnet unit has different polarities in a longitudinal direction). In addition, both side surfaces of the magnet unit in the other direction different from the one direction may be magnetized to different polarities (i.e., the magnet unit has different polarities in a width direction).
(35) The magnetic field formed by each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to the embodiments of the present disclosure is illustrated as a one-dot chain line in each drawing.
(36) The terms "left side," "right side," "upper side," "lower side," "front side," and "rear side" used in the following description will be understood based on a coordinate system illustrated in FIG. **2**.
2. Description of Configuration of Direct Current Relay **1** According to Embodiment of Present Disclosure
(37) Referring to FIGS. **2** to **4**, a direct current relay **1** according to the embodiment of the present disclosure includes a frame part **10**, an opening/closing part **20**, a core part **30**, and a movable contactor part **40**.
(38) In addition, referring to FIGS. **5** to **36**, the direct current relay **1** according to the embodiment of the present disclosure includes an arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**.
(39) The arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** may form a discharge path of a generated arc.
(40) Hereinafter, each configuration of the direct current relay **1** according to the embodiment of the present disclosure will be described with reference to the accompanying drawings, and the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** will be described as separate clauses.
(41) The description is made on the assumption that the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to various embodiments described below are each provided in the direct current relay **1**.

(42) However, it will be understood that the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** are applicable to a device in a form that can be electrically connected to and disconnected from the outside by the contact and separation between a fixed contact and a movable contact, such as a magnetic contactor, a magnetic switch, or the like.
(1) Description of Frame Part **10**
(43) The frame part **10** forms an outer side of the direct current relay **1**. A predetermined space is formed in the frame part **10**. Various devices for the direct current relay **1** to perform functions for applying or cutting off current transmitted from the outside may be accommodated in the space.
(44) That is, the frame part **10** serves as a kind of housing.
(45) The frame part **10** may be formed of an insulating material such as synthetic resin. This is for preventing an arbitrary electrical connection between the inside and outside of the frame part **10**.
(46) The frame part **10** includes an upper frame **11**, a lower frame **12**, an insulating plate **13**, and a supporting plate **14**.
(47) The upper frame **11** forms an upper side of the frame part **10**. A predetermined space is formed inside the upper frame **11**.
(48) The opening/closing part **20** and the movable contactor part **40** may be accommodated in the inner space of the upper frame **11**. The arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** may also be accommodated in the inner space of the upper frame **11**.
(49) The upper frame **11** may be coupled to the lower frame **12**. The insulating plate **13** and the supporting plate **14** may be provided in a space between the upper frame **11** and the lower frame **12**.
(50) The fixed contactor **22** of the opening/closing part **20** is located on one side of the upper frame **11**, e.g., on an upper side of the upper frame **11** in the illustrated embodiment. The fixed contactor **22** may be partially exposed to the upper side of the upper frame **11** to be electrically connected to an external power supply or a load.
(51) To this end, a through hole through which the fixed contactor **22** is coupled may be formed at the upper side of the upper frame **11**.
(52) The lower frame **12** forms a lower side of the frame part **10**. A predetermined space is formed inside the lower frame **12**. The core part **30** may be accommodated in the inner space of the lower frame **12**.
(53) The lower frame **12** may be coupled to the upper frame **11**. The insulating plate **13** and the supporting plate **14** may be provided in the space between the lower frame **12** and the upper frame **11**.
(54) The insulating plate **13** and the supporting plate **14** electrically and physically isolate the inner space of the upper frame **11** and the inner space of the lower frame **12** from each other.
(55) The insulating plate **13** is located between the upper frame **11** and the lower frame **12**. The insulating plate **13** allows the upper frame **11** and the lower frame **12** to be electrically separated from each other. To this end, the insulating plate **13** may be formed of an insulating material such as synthetic resin.
(56) Arbitrary electrical connection between the opening/closing part **20**, the movable contactor part **40**, and the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** that are accommodated in the upper frame **11** and the core part **30** accommodated in the lower frame **12** can be prevented by the insulating plate **13**.
(57) A through hole (not shown) is formed in a central part of the insulating plate **13**. A shaft **44** of the movable contactor part **40** is coupled through the through hole (not shown) to be movable in a vertical direction.
(58) The supporting plate **14** is located on a lower side of the insulating plate **13**. The insulating plate **13** may be supported by the supporting plate **14**.
(59) The supporting plate **14** is located between the upper frame **11** and the lower frame **12**.
(60) The supporting plate **14** may allow the upper frame **11** and the lower frame **12** to be physically

separated from each other. In addition, the supporting plate **14** supports the insulating plate **13**.
(61) The supporting plate **14** may be formed of a magnetic material. Accordingly, the supporting plate **14** may form a magnetic circuit together with a yoke **33** of the core part **30**. A driving force allowing a movable core **32** of the core part **30** to move toward a fixed core **31** may be formed by the magnetic circuit.
(62) A through hole (not shown) is formed in a central part of the supporting plate **14**. The shaft **44** is coupled through the through hole (not shown) to be movable in the vertical direction.
(63) Accordingly, when the movable core **32** is moved in a direction toward or away from the fixed core **31**, the shaft **44** and the movable contactor **43** connected to the shaft **44** may also be moved in the same direction.
(2) Description of Opening/Closing Part **20**
(64) The opening/closing part **20** may allow or block the flow of current according to an operation of the core part **30**. Specifically, the opening/closing part **20** may allow or block the flow of current as the fixed contactor **22** and the movable contactor **43** are brought into contact with or separated from each other.
(65) The opening/closing part **20** is accommodated in the inner space of the upper frame **11**. The opening/closing part **20** may be electrically and physically separated from the core part **30** by the insulating plate **13** and the supporting plate **14**.
(66) The opening/closing part **20** includes an arc chamber **21**, the fixed contactor **22**, and a sealing member **23**.
(67) In addition, the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** may be provided outside the arc chamber **21**. The arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** may form a magnetic field for forming an arc path A.P of an arc generated inside the arc chamber **21**. A detailed description thereof will be given below.
(68) The arc chamber **21** extinguishes an arc at an inner space thereof, wherein the arc is generated as the fixed contactor **22** and the movable contactor **43** are separated from each other. Accordingly, the arc chamber **21** may also be referred to as an “arc extinguishing part.”
(69) The arc chamber **21** sealingly accommodates the fixed contactor **22** and the movable contactor **43**. That is, the fixed contactor **22** and the movable contactor **43** are accommodated in the arc chamber **21**. Accordingly, the arc generated as the fixed contactor **22** and the movable contactor **43** are separated from each other does not arbitrarily leak to the outside.
(70) An extinguishing gas may be filled in the arc chamber **21**. The extinguishing gas may extinguish the generated arc and the extinguished arc may be discharged to the outside of the direct current relay **1** through a predetermined path. To this end, a communication hole (not shown) may be formed in a wall surrounding the inner space of the arc chamber **21**.
(71) The arc chamber **21** may be formed of an insulating material. In addition, the arc chamber **21** may be formed of a material having high pressure resistance and high heat resistance. This is because the generated arc is a flow of high-temperature and high-pressure electrons. In one embodiment, the arc chamber **21** may be formed of a ceramic material.
(72) A plurality of through holes may be formed in an upper side of the arc chamber **21**. The fixed contactor **22** is coupled through each of the through holes.
(73) In the illustrated embodiment, two fixed contactors **22** including a first fixed contactor **22***a* and a second fixed contactor **22***b* are provided. Accordingly, two through hole formed in the upper side of the arc chamber **21** may also be provided.
(74) When the fixed contactors **22** are coupled through the through holes, the through holes are sealed. That is, the fixed contactor **22** is sealingly coupled to the through hole. Accordingly, the generated arc cannot be discharged to the outside through the through hole.
(75) A lower side of the arc chamber **21** may be open. The lower side of the arc chamber **21** may be in contact with the insulating plate **13** and the sealing member **23**. That is, the lower side of the arc chamber **21** is sealed by the insulating plate **13** and the sealing member **23**.

(76) Accordingly, the arc chamber **21** can be electrically and physically separated from an outer space of the upper frame **11**.
(77) The arc extinguished in the arc chamber **21** is discharged to the outside of the direct current relay **1** through a predetermined path. In one embodiment, the extinguished arc may be discharged to the outside of the arc chamber **21** through the communication hole (not shown).
(78) The fixed contactor **22** may be brought into contact with or separated from the movable contactor **43**, so that the inside and outside of the direct current relay **1** are electrically connected or disconnected.
(79) Specifically, when the fixed contactor **22** is brought into contact with the movable contactor **43**, the inside and outside of the direct current relay **1** may be electrically connected. On the other hand, when the fixed contactor **22** is separated from the movable contactor **43**, the inside and outside of the direct current relay **1** may be electrically disconnected.
(80) As the name implies, the fixed contactor **22** does not move. That is, the fixed contactor **22** may be fixedly coupled to the upper frame **11** and the arc chamber **21**. Accordingly, the contact and separation between the fixed contactor **22** and the movable contactor **43** can be achieved by the movement of the movable contactor **43**.
(81) One end portion of the fixed contactor **22**, for example, an upper end portion of the fixed contactor **22** in the illustrated embodiment, is exposed to the outside of the upper frame **11**. A power supply or a load may each be electrically connected to the one end portion.
(82) The fixed contactor **22** may be provided in plural. In the illustrated embodiment, a total of two fixed contactors **22** are provided, including the first fixed contactor **22***a* on a left side and the second fixed contactor **22***b* on a right side.
(83) The first fixed contactor **22***a* is located to be biased to one side from a center of the movable contactor **43** in the longitudinal direction, i.e., to a left side in the illustrated embodiment. In addition, the second fixed contactor **22***b* is located to be biased to another side from the center of the movable contactor **43** in the longitudinal direction, i.e., to a right side in the illustrated embodiment.
(84) A power supply may be electrically connected to any one of the first fixed contactor **22***a* and the second fixed contactor **22***b*. In addition, a load may be electrically connected to the other one of the first fixed contactor **22***a* and the second fixed contactor **22***b*.
(85) The direct current relay **1** according to the embodiment of the present disclosure may form the arc path A.P regardless of a direction of the power supply or load connected to the fixed contactor **22**. This can be achieved by the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**, and a detailed description thereof will be described below.
(86) The other end portion of the fixed contactor **22**, i.e., a lower end portion of the fixed contactor **22** in the illustrated embodiment extends toward the movable contactor **43**.
(87) When the movable contactor **43** is moved in a direction toward the fixed contactor **22**, i.e., upward in the illustrated embodiment, the lower end portion of the fixed contactor **22** is brought into contact with the movable contactor **43**. Accordingly, the outside and inside of the direct current relay **1** can be electrically connected.
(88) The lower end portion of the fixed contactor **22** may be located inside the arc chamber **21**.
(89) When control power is cut off, the movable contactor **43** is separated from the fixed contactor **22** by an elastic force of a return spring **36**.
(90) At this time, as the fixed contactor **22** and the movable contactor **43** are separated from each other, an arc is generated between the fixed contactor **22** and the movable contactor **43**. The generated arc may be extinguished by the extinguishing gas inside the arc chamber **21**, and may be discharged to the outside along a path formed by the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**.
(91) The sealing member **23** may block the inner space of the arc chamber **21** from arbitrarily communicating with the inner space of the upper frame **11**. The sealing member **23** seals the lower

side of the arc chamber **21** together with the insulating plate **13** and the supporting plate **14**.
(92) Specifically, an upper side of the sealing member **23** is coupled to the lower side of the arc chamber **21**. In addition, a radially inner side of the sealing member **23** is coupled to an outer circumference of the insulating plate **13**, and a lower side of the sealing member **23** is coupled to the supporting plate **14**.
(93) Accordingly, the arc generated in the arc chamber **21** and the arc extinguished by the extinguishing gas do not arbitrarily flow out to the inner space of the upper frame **11**.
(94) In addition, the sealing member **23** may be configured to block an inner space of a cylinder **37** from arbitrarily communicating with the inner space of the frame part **10**.
(3) Description of Core Part **30**
(95) The core part **30** moves the movable contactor part **40** upward as control power is applied. In addition, when the application of the control power is released, the core part **30** moves the movable contactor part **40** downward again.
(96) The core part **30** may be electrically connected to an external control power supply (not shown) to receive the control power.
(97) The core part **30** is located below the opening/closing part **20**. In addition, the core part **30** is accommodated in the lower frame **12**. The core part **30** and the opening/closing part **20** may be electrically and physically separated from each other by the insulating plate **13** and the supporting plate **14**.
(98) The movable contactor part **40** is located between the core part **30** and the opening/closing part **20**. The movable contactor part **40** may be moved by a driving force applied by the core part **30**. Accordingly, the movable contactor **43** and the fixed contactor **22** can be brought into contact with each other so that current can flow through the direct current relay **1**.
(99) The core part **30** includes the fixed core **31**, the movable core **32**, the yoke **33**, a bobbin **34**, coils **35**, the return spring **36**, and the cylinder **37**.
(100) The fixed core **31** is magnetized by a magnetic field generated in the coils **35** to generate an electromagnetic attractive force. The movable core **32** is moved toward the fixed core **31** (in an upward direction in FIG. **3**) by the electromagnetic attractive force.
(101) The fixed core **31** is not moved. That is, the fixed core **31** is fixedly coupled to the supporting plate **14** and the cylinder **37**.
(102) The fixed core **31** may be provided in any form capable of being magnetized by the magnetic field so as to generate an electromagnetic force. In one embodiment, the fixed core **31** may be provided as a permanent magnet, an electromagnet, or the like.
(103) The fixed core **31** is partially accommodated in an upper space inside the cylinder **37**. In addition, an outer circumference of the fixed core **31** may come into contact with an inner circumference of the cylinder **37**.
(104) The fixed core **31** is located between the supporting plate **14** and the movable core **32**.
(105) A through hole (not shown) is formed in a central portion of the fixed core **31**. The shaft **44** is coupled through the through hole (not shown) to be movable up and down.
(106) The fixed core **31** is located to be spaced apart from the movable core **32** by a predetermined distance. Accordingly, a distance by which the movable core **32** can move toward the fixed core **31** may be limited to the predetermined distance. Accordingly, the predetermined distance may be defined as a “moving distance of the movable core **32**.”
(107) One end portion of the return spring **36**, i.e., an upper end portion of the return spring **36** in the illustrated embodiment may be brought into contact with a lower side of the fixed core **31**. When the movable core **32** is moved upward as the fixed core **31** is magnetized, the return spring **36** is compressed and stores a restoring force.
(108) Accordingly, when the application of the control power is released and the magnetization of the fixed core **31** is terminated, the movable core **32** may be returned to the lower side by the restoring force.

(109) When the control power is applied, the movable core **32** is moved toward the fixed core **31** by the electromagnetic attractive force generated by the fixed core **31**.
(110) As the movable core **32** is moved, the shaft **44** coupled to the movable core **32** is moved toward the fixed core **31**, i.e., upward in the illustrated embodiment. In addition, as the shaft **44** is moved, the movable contactor part **40** coupled to the shaft **44** is moved upward.
(111) Accordingly, the fixed contactor **22** and the movable contactor **43** may be brought into contact with each other so that the direct current relay **1** can be electrically connected to the external power supply or the load.
(112) The movable core **32** may be provided in any form capable of receiving an attractive force by an electromagnetic force. In one embodiment, the movable core **32** may be formed of a magnetic material or provided as a permanent magnet, an electromagnet, or the like.
(113) The movable core **32** is accommodated in the cylinder **37**. In addition, the movable core **32** may be moved in the cylinder **37** in the longitudinal direction of the cylinder **37**, for example, in the vertical direction in the illustrated embodiment.
(114) Specifically, the movable core **32** may be moved in a direction toward the fixed core **31** and away from the fixed core **31**.
(115) The movable core **32** is coupled to the shaft **44**. The movable core **32** may be moved integrally with the shaft **44**. When the movable core **32** is moved upward or downward, the shaft **44** is also moved upward or downward. Accordingly, the movable contactor **43** is also moved upward or downward.
(116) The movable core **32** is located below the fixed core **31**. The movable core **32** is spaced apart from the fixed core **31** by the predetermined distance. As described above, the predetermined distance is a distance by which the movable core **32** can be moved in the vertical direction.
(117) The movable core **32** is formed to extend in the longitudinal direction. A hollow part extending in the longitudinal direction is formed to be recessed in the movable core **32** by a predetermined distance. The return spring **36** and the lower side of the shaft **44** coupled through the return spring **36** are partially accommodated in the hollow part.
(118) A through hole may be formed through a lower side of the hollow part in the longitudinal direction. The hollow part and the through hole communicate with each other. A lower end portion of the shaft **44** inserted into the hollow part may proceed toward the through hole.
(119) A space part is formed to be recessed in a lower end portion of the movable core **32** by a predetermined distance. The space part communicates with the through hole. A lower head portion of the shaft **44** is located in the space part.
(120) The yoke **33** forms a magnetic circuit as the control power is applied. The magnetic circuit formed by the yoke **33** may be configured to control a direction of a magnetic field formed by the coils **35**.
(121) Accordingly, when the control power is applied, the coils **35** may form a magnetic field in a direction in which the movable core **32** is moved toward the fixed core **31**. The yoke **33** may be formed of a conductive material capable of allowing electrical connection.
(122) The yoke **33** is accommodated in the lower frame **12**. The yoke **33** surrounds the coils **35**. The coils **35** may be accommodated in the yoke **33** so as to be spaced apart from an inner circumferential surface of the yoke **33** by a predetermined distance.
(123) The bobbin **34** is accommodated in the yoke **33**. That is, the yoke **33**, the coils **35**, and the bobbin **34** on which the coils **35** are wound may be sequentially disposed in a direction from an outer circumference of the lower frame **12** toward a radially inner side of the lower frame **12**.
(124) An upper side of the yoke **33** may come into contact with the supporting plate **14**. In addition, the outer circumference of the yoke **33** may come into contact with an inner circumference of the lower frame **12** or may be located to be spaced apart from the inner circumference of the lower frame **12** by a predetermined distance.
(125) The coils **35** are wound around the bobbin **34**. The bobbin **34** is accommodated in the yoke

33.
(126) The bobbin **34** may include upper and lower portions formed in a flat plate shape, and a cylindrical column part formed to extend in the longitudinal direction to connect the upper and lower portions. That is, the bobbin **34** has a bobbin shape.
(127) The upper portion of the bobbin **34** comes into contact with a lower side of the supporting plate **14**. The coils **35** are wound around the column part of the bobbin **34**. A wound thickness of the coils **35** may be configured to be equal to or smaller than a diameter of each of the upper and lower portions of the bobbin **34**.
(128) A hollow part is formed through the column portion of the bobbin **34** extending in the longitudinal direction. The cylinder **37** may be accommodated in the hollow part. The column part of the bobbin **34** may be disposed to have the same central axis as the fixed core **31**, the movable core **32**, and the shaft **44**.
(129) The coils **35** generate a magnetic field due to the applied control power. The fixed core **31** may be magnetized by the magnetic field generated by the coils **35** and thus an electromagnetic attractive force may be applied to the movable core **32**.
(130) The coils **35** are wound around the bobbin **34**. Specifically, the coils **35** are wound around the column part of the bobbin **34** and stacked on a radial outer side of the column part. The coils **35** are accommodated in the yoke **33**.
(131) When control power is applied, the coils **35** generate a magnetic field. At this point, a strength or direction of the magnetic field generated by the coils **35** may be controlled by the yoke **33**. The fixed core **31** is magnetized by the magnetic field generated by the coils **35**.
(132) When the fixed core **31** is magnetized, the movable core **32** receives an electromagnetic force, i.e., an attractive force in a direction toward the fixed core **31**.
(133) Accordingly, the movable core **32** is moved in a direction toward the fixed core **31**, i.e., upward in the illustrated embodiment.
(134) The return spring **36** provides a restoring force for the movable core **32** to return to its original location when the application of the control power is released after the movable core **32** is moved toward the fixed core **31**.
(135) As the movable core **32** is moved toward the fixed core **31**, the return spring **36** stores the restoring force while being compressed. At this point, the stored restoring force may preferably be smaller than the electromagnetic attractive force, which is exerted on the movable core **32** as the fixed core **31** is magnetized. This is to prevent the movable core **32** from being arbitrarily returned to its original location by the return spring **36** while the control power is applied.
(136) When the application of the control power is released, the movable core **32** receives only the restoring force by the return spring **36**. Of course, gravity due to an empty weight of the movable core **32** may also be applied to the movable core **32**. Accordingly, the movable core **32** can be moved in a direction away from the fixed core **31** to be returned to the original location.
(137) The return spring **36** may be provided in any form that is deformed to store the restoring force and returned to its original state to transmit the restoring force to the outside. In one embodiment, the return spring **36** may be provided as a coil spring.
(138) The shaft **44** is coupled through the return spring **36**. The shaft **44** may move in the vertical direction regardless of the deformation of the return spring **36** in the coupled state with the return spring **36**.
(139) The return spring **36** is accommodated in the hollow part formed to be recessed in an upper side of the movable core **32**. In addition, one end portion of the return spring **36** facing the fixed core **31**, i.e., an upper end portion of the return spring **36** in the illustrated embodiment is accommodated in a hollow part formed to be recessed in the lower side of the fixed core **31**.
(140) The cylinder **37** accommodates the fixed core **31**, the movable core **32**, the return spring **36**, and the shaft **44**. The movable core **32** and the shaft **44** may be moved in the upward and downward directions in the cylinder **37**.

(141) The cylinder **37** is located in the hollow part formed in the column part of the bobbin **34**. An upper end portion of the cylinder **37** comes into contact with a lower side surface of the supporting plate **14**.
(142) A side surface of the cylinder **37** comes into contact with an inner circumferential surface of the column part of the bobbin **34**. An upper opening of the cylinder **37** may be sealed by the fixed core **31**. A lower side surface of the cylinder **37** may come into contact with an inner surface of the lower frame **12**.
(4) Description of Movable Contactor Part **40**
(143) The movable contactor part **40** includes the movable contactor **43** and components for moving the movable contactor **43**. The direct current relay **1** may be electrically connected to an external power supply or a load by the movable contactor part **40**.
(144) The movable contactor part **40** is accommodated in the inner space of the upper frame **11**. In addition, the movable contactor part **40** is accommodated in the arc chamber **21** to be movable up and down.
(145) The fixed contactor **22** is located above the movable contactor part **40**. The movable contactor part **40** is accommodated in the arc chamber **21** to be movable in a direction toward the fixed contactor **22** and a direction away from the fixed contactor **22**.
(146) The core part **30** is located below the movable contactor part **40**. The movement of the movable contactor part **40** can be achieved by the movement of the movable core **32**.
(147) The movable contactor part **40** includes a housing **41**, a cover **42**, the movable contactor **43**, the shaft **44**, and an elastic part **45**.
(148) The housing **41** accommodates the movable contactor **43** and the elastic part **45** elastically supporting the movable contactor **43**.
(149) In the illustrated embodiment, the housing **41** is formed such that one side and another side opposite to the one side are open. The movable contactor **43** may be inserted through the open portions.
(150) Unopened side surfaces of the housing **41** may be configured to surround the accommodated movable contactor **43**.
(151) The cover **42** is provided on an upper side of the housing **41**. The cover **42** covers an upper surface of the movable contactor **43** accommodated in the housing **41**.
(152) The housing **41** and the cover **42** may preferably be formed of an insulating material to prevent unexpected electrical connection. In one embodiment, the housing **41** and the cover **42** may be formed of synthetic resin or the like.
(153) A lower side of the housing **41** is connected to the shaft **44**. When the movable core **32** connected to the shaft **44** is moved upward or downward, the housing **41** and the movable contactor **43** accommodated in the housing **41** may also be moved upward or downward.
(154) The housing **41** and the cover **42** may be coupled by arbitrary members. In one embodiment, the housing **41** and the cover **42** may be coupled by coupling members (not shown) such as a bolt and a nut.
(155) The movable contactor **43** comes into contact with the fixed contactor **22** as control power is applied, so that the direct current relay **1** can be electrically connected to an external power supply and a load. In addition, when the application of the control power is released, the movable contactor **43** is separated from the fixed contactor **22**, and thus the direct current relay **1** is electrically disconnected from the external power supply and the load.
(156) The movable contactor **43** is located adjacent to the fixed contactor **22**.
(157) An upper side of the movable contactor **43** is partially covered by the cover **42**. In one embodiment, a portion of the upper surface of the movable contactor **43** may be brought into contact with a lower side surface of the cover **42**.
(158) A lower side of the movable contactor **43** is elastically supported by the elastic part **45**. In order to prevent the movable contactor **43** from being arbitrarily moved downward, the elastic part

45 may elastically support the movable contactor **43** in a compressed state by a predetermined distance.
(159) The movable contactor **43** is formed to extend in the longitudinal direction, i.e., in a left-right direction in the illustrated embodiment. That is, a length of the movable contactor **43** is formed to be longer than a width thereof. Accordingly, both end portions of the movable contactor **43** in the longitudinal direction, which are accommodated in the housing **41**, are exposed to the outside of the housing **41**.
(160) Contact protrusions may be formed to protrude upward from the both end portions by predetermined distances. The fixed contactor **22** is in contact with the contact protrusions.
(161) The contact protrusions may be formed at locations corresponding to the fixed contactors **22***a* and **22***b*, respectively. Accordingly, the moving distance of the movable contactor **43** can be reduced and contact reliability between the fixed contactor **22** and the movable contactor **43** can be improved.
(162) The width of the movable contactor **43** may be the same as a spaced distance between the side surfaces of the housing **41**. That is, when the movable contactor **43** is accommodated in the housing **41**, both side surfaces of the movable contactor **43** in a width direction may be brought into contact with inner surfaces of the side surfaces of the housing **41**.
(163) Accordingly, the state in which the movable contactor **43** is accommodated in the housing **41** can be stably maintained.
(164) The shaft **44** transmits a driving force, which is generated in response to the operation of the core part **30**, to the movable contactor part **40**. Specifically, the shaft **44** is connected to the movable core **32** and the movable contactor **43**. When the movable core **32** is moved upward or downward, the movable contactor **43** may also be moved upward or downward by the shaft **44**.
(165) The shaft **44** is formed to extend in the longitudinal direction, i.e., in the vertical direction in the illustrated embodiment.
(166) The lower end portion of the shaft **44** is inserted into and coupled to the movable core **32**. When the movable core **32** is moved in the vertical direction, the shaft **44** may also be moved in the vertical direction together with the movable core **32**.
(167) A body part of the shaft **44** is coupled through the fixed core **31** to be movable up and down. The return spring **36** is coupled through the body part of the shaft **44**.
(168) An upper end portion of the shaft **44** is coupled to the housing **41**. When the movable core **32** is moved, the shaft **44** and the housing **41** may also be moved together with the movable core **32**.
(169) The upper and lower end portions of the shaft **44** may be formed to have a larger diameter than the body part of the shaft. Accordingly, the coupled state of the shaft **44** to the housing **41** and the movable core **32** can be stably maintained.
(170) The elastic part **45** elastically supports the movable contactor **43**. When the movable contactor **43** is brought into contact with the fixed contactor **22**, the movable contactor **43** may tend to be separated from the fixed contactor **22** due to an electromagnetic repulsive force.
(171) At this point, the elastic part **45** elastically supports the movable contactor **43** to prevent the movable contactor **43** from being arbitrarily separated from the fixed contactor **22**.
(172) The elastic part **45** may be provided in any form capable of storing a restoring force by being deformed and providing the stored restoring force to another member. In one embodiment, the elastic part **45** may be provided as a coil spring.
(173) One end portion of the elastic part **45** facing the movable contactor **43** comes into contact with the lower side of the movable contactor **43**. In addition, the other end portion opposite to the one end portion comes into contact with the upper side of the housing **41**.
(174) The elastic part **45** may elastically support the movable contactor **43** in a state of storing the restoring force by being compressed by a predetermined distance. Accordingly, even when the electromagnetic repulsive force is generated between the movable contactor **43** and the fixed contactor **22**, the movable contactor **43** is not arbitrarily moved.

(175) A protrusion (not shown) inserted into the elastic part **45** may be formed to protrude from the lower side of the movable contactor **43** to enable stable coupling of the elastic part **45**. Similarly, a protrusion (not shown) inserted into the elastic part **45** may also be formed to protrude from the upper side of the housing **41**.

3. Description of Arc Path Formation Units **100**, **200**, **300**, **400**, **500**, and **600** According to Embodiments of Present Disclosure

(176) Referring to FIGS. **5** to **36**, the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to various embodiments of the present disclosure are illustrated. Each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** forms magnetic fields inside the arc chamber **21**. Due to current flowing through the direct current relay **1** and the formed magnetic field, an electromagnetic force is formed in the arc chamber **21**.

(177) An arc generated as the fixed contactor **22** and the movable contactor **43** are separated from each other is moved to the outside of the arc chamber **21** by the formed electromagnetic force. Specifically, the generated arc is moved in a direction of the formed electromagnetic force. Accordingly, it can be said that each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** forms an arc path A.P, which is a path through which the generated arc flows.

(178) Each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** is located in a space formed in the upper frame **11**. The arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600** is disposed to surround the arc chamber **21**. In other words, the arc chamber **21** is located inside the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**.

(179) The fixed contactor **22** and the movable contactor **43** are located inside the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**. The arc generated as the fixed contactor **22** and the movable contactor **43** are separated from each other may be induced by the electromagnetic force formed by the arc path formation unit **100**, **200**, **300**, **400**, **500**, or **600**.

(180) Each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to various embodiments of the present disclosure includes Halbach arrays or magnet units. The Halbach arrays or the magnet units form magnetic fields inside the arc path formation unit **100** in which the fixed contactor **22** and the movable contactor **43** are accommodated. At this point, the Halbach array or the magnet part may form a magnetic field by itself and between each other.

(181) The magnetic fields formed by the Halbach array and the magnet unit form an electromagnetic force together with current flowing through the fixed contactor **22** and the movable contactor **43**. The formed electromagnetic force induces an arc that is generated when the fixed contactor **22** and the movable contactor **43** are separated from each other.

(182) At this point, each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** forms the electromagnetic force in a direction away from a central part C of each of space parts **115**, **215**, **315**, **415**, **515**, and **615**. Accordingly, an arc path A.P is also formed in the direction away from a central part C of the space part.

(183) As a result, each component provided in the direct current relay **1** is not damaged by the generated arc. Furthermore, the generated arc may be quickly discharged to the outside of the arc chamber **21**.

(184) Hereinafter, the configuration of each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** and the arc path A.P formed by each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** will be described in detail with reference to the accompanying drawings.

(185) Each of the arc path formation units **100**, **200**, **300**, **400**, **500**, and **600** according to various embodiments described below may include the Halbach array located on one or more of front and rear sides to be biased to any one side of left and right sides.

(186) That is, the Halbach array may be disposed adjacent to any one of the first fixed contactor **22***a* located on the left side and the second fixed contactor **22***b* located on the right side.

(187) As will be described below, the rear side may be defined as a direction adjacent to a first surface **111**, **211**, **311**, **411**, **511**, or **611** and the front side may be defined as a direction adjacent to a

second surface **112**, **212**, **312**, **412**, **512**, or **612**.
(188) In addition, the left side may be defined as a direction adjacent to a third surface **113**, **213**, **313**, **413**, **513**, or **613**, and the right side may be defined as a direction adjacent to a fourth surface **114**, **214**, **314**, **414**, **514**, or **614**.
(1) Description of Arc Path Formation Unit **100** According to One Embodiment of Present Disclosure
(189) Hereinafter, an arc path formation unit **100** according to one embodiment of the present disclosure will be described in detail with reference to FIGS. **5** to **8**.
(190) Referring to FIGS. **5** and **6**, the arc path formation unit **100** according to the illustrated embodiment includes a magnet frame **110**, a first Halbach array **120**, a second Halbach array **130**, and a third Halbach array **140**.
(191) The magnet frame **110** forms a frame of the arc path formation unit **100**. The first Halbach array **120**, the second Halbach array **130**, and the third Halbach array **140** are disposed in the magnet frame **110**. In one embodiment, the first Halbach array **120**, the second Halbach array **130**, and the third Halbach array **140** may be coupled to the magnet frame **110**.
(192) The magnet frame **110** has a rectangular cross section formed to extend in the longitudinal direction, i.e., in the left-right direction in the illustrated embodiment. The shape of the magnet frame **110** may be changed depending on shapes of the upper frame **11** and the arc chamber **21**.
(193) The magnet frame **110** includes a first surface **111**, a second surface **112**, a third surface **113**, a fourth surface **114**, and a space part **115**.
(194) The first surface **111**, the second surface **112**, the third surface **113**, and the fourth surface **114** form an outer circumferential surface of the magnet frame **110**. That is, the first surface **111**, the second surface **112**, the third surface **113**, and the fourth surface **114** may serve as walls of the magnet frame **110**.
(195) An outer side of each of the first surface **111**, the second surface **112**, the third surface **113**, and the fourth surface **114** may be in contact with or fixedly coupled to an inner surface of the upper frame **11**. In addition, the first Halbach array **120**, the second Halbach array **130**, and the third Halbach array **140** may be located on inner sides of the first surface **111**, the second surface **112**, the third surface **113**, and the fourth surface **114**.
(196) In the illustrated embodiment, the first surface **111** forms a rear side surface. The second surface **112** forms a front side surface and faces the first surface **111**. In addition, the third surface **113** forms a left side surface. The fourth surface **114** forms a right side surface and faces the third surface **113**.
(197) That is, the first surface **111** and the second surface **112** face each other with the space part **115** therebetween. In addition, the third surface **113** and the fourth surface **114** face each other with the space part **115** therebetween.
(198) The first surface **111** is continuous with the third surface **113** and the fourth surface **114**. The first surface **111** may be coupled to the third surface **113** and the fourth surface **114** at predetermined angles. In one embodiment, the predetermined angle may be a right angle.
(199) The second surface **112** is continuous with the third surface **113** and the fourth surface **114**. The second surface **112** may be coupled to the third surface **113** and the fourth surface **114** at predetermined angles. In one embodiment, the predetermined angle may be a right angle.
(200) Each of corners at which the first to fourth surfaces **111** to **114** are connected to each other may be chamfered.
(201) Coupling members (not shown) may be provided to couple the first to third Halbach arrays **120**, **130**, and **140** to the respective surfaces **111**, **112**, **113**, and **114**.
(202) Although not illustrated in the drawings, an arc discharge hole (not shown) may be formed through one or more of the first surface **111**, the second surface **112**, the third surface **113**, and the fourth surface **114**. The arc discharge hole (not shown) may serve as a path through which an arc generated in the space part **115** is discharged.

(203) A space surrounded by the first to fourth surfaces **111** to **114** may be defined as the space part **115**.
(204) The fixed contactor **22** and the movable contactor **43** are accommodated in the space part **115**. In addition, the arc chamber **21** is accommodated in the space part **115**.
(205) In the space part **115**, the movable contactor **43** may be moved in a direction toward the fixed contactor **22** (i.e., the downward direction) or a direction away from the fixed contactor **22** (i.e., the upward direction).
(206) In addition, an arc path A.P of an arc generated in the arc chamber **21** is formed in the space part **115**. This is achieved by the magnetic field formed by the first to third Halbach arrays **120**, **130**, and **140**.
(207) A central portion of the space part **115** may be defined as the central part C. A straight line distance from each of corners at which the first to fourth surfaces **111** to **114** are connected to each other to the central part C may be formed to be equal to each other.
(208) The central part C may be located between the first fixed contactor **22***a* and the second fixed contactor **22***b*. In addition, a central portion of the movable contactor part **40** is located vertically below the central part C. That is, a central portion of each of the housing **41**, the cover **42**, the movable contactor **43**, the shaft **44**, the elastic part **45**, and the like is located vertically below the central part C.
(209) Accordingly, when the generated arc is moved toward the central part C, the above components may be damaged. In order to prevent this, the arc path formation unit **100** according to the present embodiment includes the first Halbach array **120**, the second Halbach array **130**, and the third Halbach array **140**.
(210) In the illustrated embodiment, the plurality of magnetic materials constituting the first Halbach array **120** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the first Halbach array **120** is formed to extend in the left-right direction.
(211) The first Halbach array **120** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the first Halbach array **120** may form magnetic fields together with the second and third Halbach arrays **130** and **140**.
(212) The first Halbach array **120** may be located adjacent to any one surface of the first and second surfaces **111** and **112**. In one embodiment, the first Halbach array **120** may be coupled to an inner side (i.e., the side in a direction toward the space part **115**) of the any one surface.
(213) In the illustrated embodiment, the first Halbach array **120** is disposed on the inner side of the first surface **111** and adjacent to the first surface **111** and faces the second Halbach array **130** located on the inner side of the second surface **112**.
(214) The space part **115**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **115** are located between the first Halbach array **120** and the second Halbach array **130**.
(215) In addition, the space part **115**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **115** are located between the first Halbach array **120** and the third Halbach array **140**.
(216) The first Halbach array **120** may be located to be biased to any one surface of the third surface **113** and the fourth surface **114**. In the embodiment illustrated in FIG. **5**, the first Halbach array **120** is located to be biased to the third surface **113**. In the embodiment illustrated in FIG. **6**, the first Halbach array **120** is located to be biased to the fourth surface **114**.
(217) The first Halbach array **120** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the second Halbach array **130** and the third Halbach array **140**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first Halbach array **120** is a well-known technique, a detailed description thereof will be omitted.

(218) In the illustrated embodiment, the first Halbach array **120** includes a first block **121**, a second block **122**, and a third block **123**. It will be understood that the plurality of magnetic materials constituting the first Halbach array **120** are named as the blocks **121**, **122**, and **123**, respectively.
(219) The first to third blocks **121**, **122**, and **123** may each be formed of a magnetic material. In one embodiment, the first to third blocks **121**, **122**, and **123** may each be provided as a permanent magnet, an electromagnet, or the like.
(220) The first to third blocks **121**, **122**, and **123** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **121**, **122**, and **123** are disposed in parallel in a direction in which the first surface **111** extends, that is, in the left-right direction.
(221) The first block **121** is located adjacent to the any one surface of the third surface **113** and the fourth surface **114**. In addition, the third block **123** is located adjacent to the other surface of the third surface **113** and the fourth surface **114**. The second block **122** is located between the first block **121** and the third block **123**.
(222) In one embodiment, the blocks **121**, **122**, and **123** adjacent to each other may be in contact with each other.
(223) The second block **122** may be disposed to overlap a second block **132** of the second Halbach array **130** and any one of the fixed contactors **22*a*** and **22*b*** in a direction toward the second Halbach array **130** or the space part **115**, i.e., in the front-rear direction in the illustrated embodiment.
(224) Each of the blocks **121**, **122**, and **123** includes a plurality of surfaces.
(225) Specifically, the first block **121** includes a first inner surface **121*a*** facing the second block **122** and a first outer surface **121*b*** opposite to the second block **122**.
(226) The second block **122** includes a second inner surface **122*a*** facing the space part **115** or the second Halbach array **130** and a second outer surface **122*b*** opposite to the space part **115** or the second Halbach array **130**.
(227) The third block **123** includes a third inner surface **123*a*** facing the second block **122** and a third outer surface **123*b*** opposite to the second block **122**.
(228) The plurality of surfaces of each of the blocks **121**, **122**, and **123** may be magnetized according to a predetermined rule to configure a Halbach array.
(229) Specifically, the first to third inner surfaces **121*a***, **122*a***, and **123*a*** may be magnetized to the same polarity. Similarly, the first to third outer surfaces **121*b***, **122*b***, and **123*b*** may be magnetized to a polarity different from the polarity of the first to third inner surfaces **121*a***, **122*a***, and **123*a***.
(230) At this point, the first to third inner surfaces **121*a***, **122*a***, and **123*a*** may be magnetized to the same polarity as first to third inner surfaces **131*a***, **132*a***, and **133*a*** of the second Halbach array **130** and first to third inner surfaces **141*a***, **142*a***, and **143*a*** of the third Halbach array **140**.
(231) Similarly, the first to third outer surfaces **121*b***, **122*b***, and **123*b*** may be magnetized to the same polarity as first to third outer surfaces **131*b***, **132*b***, and **133*b*** of the second Halbach array **130** and first to third outer surfaces **141*b***, **142*b***, and **143*b*** of the third Halbach array **140**.
(232) In the illustrated embodiment, the plurality of magnetic materials constituting the second Halbach array **130** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the second Halbach array **130** is formed to extend in the left-right direction.
(233) The second Halbach array **130** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the second Halbach array **130** may form magnetic fields together with the first and third Halbach arrays **120** and **140**.
(234) The second Halbach array **130** may be located adjacent to the other surface of the first and second surfaces **111** and **112**. In one embodiment, the second Halbach array **130** may be coupled to the inner side (i.e., the side in a direction toward the space part **115**) of the any one surface.
(235) In the illustrated embodiment, the second Halbach array **130** is disposed on the inner side of the second surface **112** and adjacent to the second surface **112** and faces the first Halbach array **120** located on the inner side of the first surface **111**.

(236) The space part **115**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **115** are located between the second Halbach array **130** and the first Halbach array **120**.
(237) In addition, the space part **115**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **115** are located between the second Halbach array **130** and the third Halbach array **140**.
(238) The second Halbach array **130** may be located to be biased to the any one surface of the third surface **113** and the fourth surface **114**. In the embodiment illustrated in FIG. **5**, the second Halbach array **130** is located to be biased to the third surface **113**. In the embodiment illustrated in FIG. **6**, the second Halbach array **130** is located to be biased to the fourth surface **114**.
(239) The second Halbach array **130** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first Halbach array **120** and the third Halbach array **140**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second Halbach array **130** is a well-known technique, a detailed description thereof will be omitted.
(240) In the illustrated embodiment, the second Halbach array **130** includes a first block **131**, the second block **132**, and a third block **133**. It will be understood that the plurality of magnetic materials constituting the second Halbach array **130** are named as the blocks **131**, **132**, and **133**, respectively.
(241) The first to third blocks **131**, **132**, and **133** may each be formed of a magnetic material. In one embodiment, the first to third blocks **131**, **132**, and **133** may each be provided as a permanent magnet, an electromagnet, or the like.
(242) The first to third blocks **131**, **132**, and **133** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **131**, **132**, and **133** are disposed in parallel in a direction in which the second surface **112** extends, that is, in the left-right direction.
(243) The first block **131** is located adjacent to the any one surface of the third surface **113** and the fourth surface **114**. In addition, the third block **133** is located adjacent to the other surface of the third surface **113** and the fourth surface **114**. The second block **132** is located between the first block **131** and the third block **133**.
(244) In one embodiment, the blocks **131**, **132**, and **133** adjacent to each other may be in contact with each other.
(245) The second block **132** may be disposed to overlap the second block **122** of the first Halbach array **120** and any one of the fixed contactors **22**_a_ and **22**_b_ in a direction toward the first Halbach array **120** or the space part **115**, i.e., in the front-rear direction in the illustrated embodiment.
(246) Each of the blocks **131**, **132**, and **133** includes a plurality of surfaces.
(247) Specifically, the first block **131** includes the first inner surface **131**_a_ facing the second block **132** and the first outer surface **131**_b_ opposite to the second block **132**.
(248) The second block **132** includes the second inner surface **132**_a_ facing the space part **115** or the first Halbach array **120**, and the second outer surface **132**_b_ opposite to the space part **115** or the first Halbach array **120**.
(249) The third block **133** includes the third inner surface **133**_a_ facing the second block **132** and the third outer surface **133**_b_ opposite to the second block **132**.
(250) The plurality of surfaces of each of the blocks **131**, **132**, and **133** may be magnetized according to a predetermined rule to configure a Halbach array.
(251) Specifically, the first to third inner surfaces **131**_a_, **132**_a_, and **133**_a_ may be magnetized to the same polarity. Similarly, the first to third outer surfaces **131**_b_, **132**_b_, and **133**_b_ may be magnetized to a polarity different from the polarity of the first to third inner surfaces **131**_a_, **132**_a_, and **133**_a_.
(252) At this point, the first to third inner surfaces **131**_a_, **132**_a_, and **133**_a_ may be magnetized to the same polarity as the first to third inner surfaces **121**_a_, **122**_a_, and **123**_a_ of the first Halbach array **120** and the first to third inner surfaces **141**_a_, **142**_a_, and **143**_a_ of the third Halbach array **140**.

(253) Similarly, the first to third outer surfaces **131***b*, **132***b*, and **133***b* may be magnetized to the same polarity as the first to third outer surfaces **121***b*, **122***b*, and **123***b* of the first Halbach array **120** and the first to third outer surfaces **141***b*, **142***b*, and **143***b* of the third Halbach array **140**.
(254) In the illustrated embodiment, the plurality of magnetic materials constituting the third Halbach array **140** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the third Halbach array **140** is formed to extend in the left-right direction.
(255) The third Halbach array **140** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the third Halbach array **140** may form magnetic fields together with the first and second Halbach arrays **120** and **130**.
(256) The third Halbach array **140** may be located adjacent to the other surface of the third and fourth surfaces **113** and **114**. In one embodiment, the first third Halbach array **140** may be coupled to the inner side (i.e., the side in a direction toward the space part **115**) of the any one surface.
(257) At this point, the third Halbach array **140** is located on a surface opposite to any one surface of the third and fourth surfaces **113** and **114**, to which the first and second Halbach arrays **120** and **130** are located to be biased.
(258) In the embodiment illustrated in FIG. **5**, the third Halbach array **140** is located adjacent to the fourth surface **114**. At this point, the first and second Halbach arrays **120** and **130** are located to be biased to the third surface **113**.
(259) In the embodiment illustrated in FIG. **6**, the third Halbach array **140** is located adjacent to the third surface **113**. At this point, the first and second Halbach arrays **120** and **130** are located to be biased to the fourth surface **114**.
(260) The third Halbach array **140** is disposed to face the other surface of the third and fourth surfaces **113** and **114**, to which the third Halbach array **140** is not adjacent. The space part **115**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **115** are located between the third Halbach array **140** and the other surface.
(261) The third Halbach array **140** is located between the first surface **111** and the second surface **112**. In one embodiment, the third Halbach array **140** may be located at a central portion of the other surface of the third and fourth surfaces **113** and **114**.
(262) In other words, the shortest distance between the third Halbach array **140** and the first surface **111** and the shortest distance between the third Halbach array **140** and the second surface **112** may be the same.
(263) The third Halbach array **140** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first Halbach array **120** and the second Halbach array **130**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the third Halbach array **140** is a well-known technique, a detailed description thereof will be omitted.
(264) In the illustrated embodiment, the third Halbach array **140** includes a first block **141**, a second block **142**, and a third block **143**. It will be understood that the plurality of magnetic materials constituting the third Halbach array **140** are named as the blocks **141**, **142**, and **143**, respectively.
(265) The first to third blocks **141**, **142**, and **143** may each be formed of a magnetic material. In one embodiment, the first to third blocks **141**, **142**, and **143** may each be provided as a permanent magnet, an electromagnet, or the like.
(266) The first to third blocks **141**, **142**, and **143** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **141**, **142**, and **143** are disposed in parallel in a direction in which the third surface **113** or the fourth surface **114** extends, that is, in the front-rear direction.
(267) The first block **141** is located at the most rear side. That is, the first block **141** is located adjacent to the first surface **111**. In addition, the third block **143** is located at the most front side.

That is, the third block **143** is located adjacent to the second surface **112**. The second block **142** is located between the first block **141** and the third block **143**.
(268) In one embodiment, the blocks **141**, **142**, and **143** adjacent to each other may be in contact with each other.
(269) The second block **142** may be disposed to overlap the fixed contactors **22***a* and **22***b* in a direction toward the space part **115**, i.e., in the left-right direction in the illustrated embodiment.
(270) Each of the blocks **141**, **142**, and **143** includes a plurality of surfaces.
(271) Specifically, the first block **141** includes the first inner surface **141***a* facing the second block **142** and the first outer surface **141***b* opposite to the second block **142**.
(272) The second block **142** includes the second inner surface **142***a* facing the space part **115** and the second outer surface **142***b* opposite to the space part **115**.
(273) The third block **143** includes the third inner surface **143***a* facing the second block **142** and the third outer surface **143***b* opposite to the second block **142**.
(274) The plurality of surfaces of each of the blocks **141**, **142**, and **143** may be magnetized according to a predetermined rule to configure a Halbach array.
(275) Specifically, the first to third inner surfaces **141***a*, **142***a*, and **143***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **141***b*, **142***b*, and **143***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **141***a*, **142***a*, and **143***a*.
(276) At this point, the first to third inner surfaces **141***a*, **142***a*, and **143***a* may be magnetized to the same polarity as the first to third inner surfaces **121***a*, **122***a*, and **123***a* of the first Halbach array **120** and the first to third inner surfaces **131***a*, **132***a*, and **133***a* of the second Halbach array **130**.
(277) Similarly, the first to third outer surfaces **141***b*, **142***b*, and **143***b* may be magnetized to the same polarity as the first to third outer surfaces **121***b*, **122***b*, and **123***b* of the first Halbach array **120** and the first to third outer surfaces **131***b*, **132***b*, and **133***b* of the second Halbach array **130**.
(278) Hereinafter, the arc path A.P formed by the arc path formation unit **100** according to the present embodiment will be described in detail with reference to FIGS. **7** and **8**.
(279) Referring to FIGS. **7** and **8**, the first to third inner surfaces **121***a*, **122***a*, and **123***a* of the first Halbach array **120** are magnetized to N poles. According to the above-described rule, the first to third inner surfaces **131***a*, **132***a*, and **133***a* of the second Halbach array **130** and the first to third inner surfaces **141***a*, **142***a*, and **143***a* of the third Halbach array **140** are also magnetized to N poles.
(280) Accordingly, magnetic fields that repel each other are formed between the first Halbach array **120** and the second Halbach array **130**. In addition, in the third Halbach array **140**, a magnetic field diverging in a direction from the second inner surface **142***a* toward the space part **115** is formed.
(281) Accordingly, in the embodiment illustrated in FIG. **7**, the magnetic fields formed by the first to third Halbach arrays **120**, **130**, and **140** are formed toward the third surface **113**, i.e., toward the left side in the illustrated embodiment.
(282) In addition, in the embodiment illustrated in FIG. **8**, the magnetic fields formed by the first to third Halbach arrays **120**, **130**, and **140** are formed toward the fourth surface **114**, i.e., toward the right side in the illustrated embodiment.
(283) In the embodiment illustrated in FIG. **7**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(284) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward a front left side.
(285) Accordingly, an arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(286) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward a rear right side.
(287) Accordingly, an arc path A.P generated in the vicinity of the second fixed contactor **22***b* is

also formed toward the rear right side.
(288) In the embodiment illustrated in FIG. **7**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(289) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward a rear left side.
(290) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(291) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward a front right side.
(292) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(293) In the embodiment illustrated in FIG. **8**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(294) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(295) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(296) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(297) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(298) In the embodiment illustrated in FIG. **8**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(299) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(300) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(301) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(302) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(303) Although not illustrated in the drawing, when the polarity of each surface of the first to third Halbach arrays **120**, **130**, and **140** is changed, the direction of the magnetic field formed in each of first to third Halbach arrays **120**, **130**, and **140** is reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(304) That is, in the electric connection situation shown in FIG. **7**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(305) Similarly, in the electric connection situation shown in FIG. **7**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.

(306) In addition, in the electric connection situation shown in FIG. **8**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(307) Similarly, in the electric connection situation shown in FIG. **8**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(308) Accordingly, in the arc path formation unit **100** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the first to third Halbach arrays **120**, **130**, and **140** or the direction of the current flowing through the direct current relay **1**.
(309) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.
(2) Description of Arc Path Formation Unit **200** According to Another Embodiment of Present Disclosure
(310) Hereinafter, an arc path formation unit **200** according to another embodiment of the present disclosure will be described in detail with reference to FIGS. **9** to **12**.
(311) Referring to FIGS. **9** and **10**, the arc path formation unit **200** according to the illustrated embodiment includes a magnet frame **210**, a first Halbach array **220**, a second Halbach array **230**, and a magnet unit **240**.
(312) The magnet frame **210** according to the present embodiment has the same structure and function as the magnet frame **110** according to the above-described embodiment. However, there is a difference in the arrangement method of the first and second Halbach arrays **220** and **230** and the magnet unit **240** disposed in the magnet frame **210** according to the present embodiment.
(313) Accordingly, a description of the magnet frame **210** will be replaced with the description of the magnet frame **110** according to the above-described embodiment.
(314) In the illustrated embodiment, the plurality of magnetic materials constituting the first Halbach array **220** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the first Halbach array **220** is formed to extend in the left-right direction.
(315) The first Halbach array **220** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the first Halbach array **220** may form magnetic fields together with the second Halbach array **230** and the magnet unit **240**.
(316) The first Halbach array **220** may be located adjacent to any one surface of first and second surfaces **211** and **212**. In one embodiment, the first Halbach array **220** may be coupled to an inner side (i.e., the side in a direction toward a space part **215**) of the any one surface.
(317) In the illustrated embodiment, the first Halbach array **220** is disposed on an inner side of the first surface **211** and adjacent to the first surface **211** and faces the second Halbach array **230** located on an inner side of the second surface **212**.
(318) The space part **215**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **215** are located between the first Halbach array **220** and the second Halbach array **230**.
(319) In addition, the space part **215**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **215** are located between the first Halbach array **220** and the magnet unit **240**.
(320) The first Halbach array **220** may be located to be biased to any one surface of a third surface **213** and a fourth surface **214**. In the embodiment illustrated in FIG. **9**, the first Halbach array **220** is located to be biased to the third surface **213**. In the embodiment illustrated in FIG. **10**, the first

Halbach array **220** is located to be biased to the fourth surface **214**.
(321) The first Halbach array **220** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the second Halbach array **230** and the magnet unit **240**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first Halbach array **220** is a well-known technique, a detailed description thereof will be omitted.
(322) In the illustrated embodiment, the first Halbach array **220** includes a first block **221**, a second block **222**, and a third block **223**. It will be understood that the plurality of magnetic materials constituting the first Halbach array **220** are named as the blocks **221**, **222**, and **223**, respectively.
(323) The first to third blocks **221**, **222**, and **223** may each be formed of a magnetic material. In one embodiment, the first to third blocks **221**, **222**, and **223** may each be provided as a permanent magnet, an electromagnet, or the like.
(324) The first to third blocks **221**, **222**, and **223** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **221**, **222**, and **223** are disposed in parallel in a direction in which the first surface **211** extends, that is, in the left-right direction.
(325) The first block **221** is located adjacent to the any one surface of the third surface **213** and the fourth surface **214**. In addition, the third block **223** is located adjacent to the other surface of the third surface **213** and the fourth surface **214**. The second block **222** is located between the first block **221** and the third block **223**.
(326) In one embodiment, the blocks **221**, **222**, and **223** adjacent to each other may be in contact with each other.
(327) The second block **222** may be disposed to overlap a second block **232** of the second Halbach array **230** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the second Halbach array **230** or the space part **215**, i.e., in the front-rear direction in the illustrated embodiment.
(328) Each of the blocks **221**, **222**, and **223** includes a plurality of surfaces.
(329) Specifically, the first block **221** includes a first inner surface **221***a* facing the second block **222** and a first outer surface **221***b* opposite to the second block **222**.
(330) The second block **222** includes a second inner surface **222***a* facing the space part **215** or the second Halbach array **230** and a second outer surface **222***b* opposite to the space part **215** or the second Halbach array **230**.
(331) The third block **223** includes a third inner surface **223***a* facing the second block **222** and a third outer surface **223***b* opposite to the second block **222**.
(332) The plurality of surfaces of each of the blocks **221**, **222**, and **223** may be magnetized according to a predetermined rule to configure a Halbach array.
(333) Specifically, the first to third inner surfaces **221***a*, **222***a*, and **223***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **221***b*, **222***b*, and **223***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **221***a*, **222***a*, and **223***a*.
(334) At this point, the first to third inner surfaces **221***a*, **222***a*, and **223***a* may be magnetized to the same polarity as first to third inner surfaces **231***a*, **232***a*, and **233***a* of the second Halbach array **230** and a facing surface **241** of the magnet unit **240**.
(335) Similarly, the first to third outer surfaces **221***b*, **222***b*, and **223***b* may be magnetized to the same polarity as first to third outer surfaces **231***b*, **232***b*, and **233***b* of the second Halbach array **230** and an opposing surface **242** of the magnet unit **240**.
(336) In the illustrated embodiment, the plurality of magnetic materials constituting the second Halbach array **230** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the second Halbach array **230** is formed to extend in the left-right direction.
(337) The second Halbach array **230** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the second Halbach array **230** may form magnetic fields together with the first Halbach array **220** and the magnet unit **240**.

(338) The second Halbach array **230** may be located adjacent to the other surface of the first and second surfaces **211** and **212**. In one embodiment, the second Halbach array **230** may be coupled to an inner side (i.e., the side in a direction toward the space part **215**) of the any one surface.
(339) In the illustrated embodiment, the second Halbach array **230** is disposed on the inner side of the second surface **212** and adjacent to the second surface **212** and faces the first Halbach array **220** located on the inner side of the first surface **211**. □j ixed contactor **22** and the movable contactor **43** accommodated in the space part **215** are located between the second Halbach array **230** and the first Halbach array **220**.
(340) In addition, the space part **215**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **215** are located between the second Halbach array **230** and the magnet unit **240**.
(341) The second Halbach array **230** may be located to be biased to the any one surface of the third surface **213** and the fourth surface **214**. In the embodiment illustrated in FIG. **9**, the second Halbach array **230** is located to be biased to the third surface **213**. In the embodiment illustrated in FIG. **10**, the second Halbach array **230** is located to be biased to the fourth surface **214**.
(342) The second Halbach array **230** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first Halbach array **220** and the magnet unit **240**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second Halbach array **230** is a well-known technique, a detailed description thereof will be omitted.
(343) In the illustrated embodiment, the second Halbach array **230** includes a first block **231**, the second block **232**, and a third block **233**. It will be understood that the plurality of magnetic materials constituting the second Halbach array **230** are named as the blocks **231**, **232**, and **233**, respectively.
(344) The first to third blocks **231**, **232**, and **233** may each be formed of a magnetic material. In one embodiment, the first to third blocks **231**, **232**, and **233** may each be provided as a permanent magnet, an electromagnet, or the like.
(345) The first to third blocks **231**, **232**, and **233** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **231**, **232**, and **233** are disposed in parallel in a direction in which the second surface **212** extends, that is, in the left-right direction.
(346) The first block **231** is located adjacent to the any one surface of the third surface **213** and the fourth surface **214**. In addition, the third block **233** is located adjacent to the other surface of the third surface **213** and the fourth surface **214**. The second block **232** is located between the first block **231** and the third block **233**.
(347) In one embodiment, the blocks **231**, **232**, and **233** adjacent to each other may be in contact with each other.
(348) The second block **232** may be disposed to overlap the second block **222** of the first Halbach array **220** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the first Halbach array **220** or the space part **215**, i.e., in the front-rear direction in the illustrated embodiment.
(349) Each of the blocks **231**, **232**, and **233** includes a plurality of surfaces.
(350) Specifically, the first block **231** includes the first inner surface **231***a* facing the second block **232** and the first outer surface **231***b* opposite to the second block **232**.
(351) The second block **232** includes the second inner surface **232***a* facing the space part **215** or the first Halbach array **220**, and the second outer surface **232***b* opposite to the space part **215** or the first Halbach array **220**.
(352) The third block **233** includes the third inner surface **233***a* facing the second block **232** and the third outer surface **233***b* opposite to the second block **232**.
(353) The plurality of surfaces of each of the blocks **231**, **232**, and **233** may be magnetized according to a predetermined rule to configure a Halbach array.
(354) Specifically, the first to third inner surfaces **231***a*, **232***a*, and **233***a* may be magnetized to the

same polarity. Similarly, the first to third outer surfaces **231***b*, **232***b*, and **233***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **231***a*, **232***a*, and **233***a*.
(355) At this point, the first to third inner surfaces **231***a*, **232***a*, and **233***a* may be magnetized to the same polarity as the first to third inner surfaces **221***a*, **222***a*, and **223***a* of the first Halbach array **220** and the facing surface **241** of the magnet unit **240**.
(356) Similarly, the first to third outer surfaces **231***b*, **232***b*, and **233***b* may be magnetized to the same polarity as the first to third outer surfaces **221***b*, **222***b*, and **223***b* of the first Halbach array **220** and the opposing surface **242** of the magnet unit **240**.
(357) The magnet unit **240** forms a magnetic field by itself, or forms magnetic fields together with the first and second Halbach arrays **220** and **230**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the magnet unit **240**.
(358) The magnet unit **240** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the magnet unit **240** may be provided as a permanent magnet, an electromagnet, or the like.
(359) The magnet unit **240** may be located adjacent to the other surface of the third and fourth surfaces **213** and **214**. In one embodiment, the magnet unit **240** may be coupled to an inner side (i.e., the side in a direction toward the space part **215**) of the other surface.
(360) At this point, the magnet unit **240** is located on a surface opposite to any one surface of the third and fourth surfaces **213** and **214**, to which the first and second Halbach arrays **220** and **230** are located to be biased.
(361) The magnet unit **240** is disposed to face the other surface of the third and fourth surfaces **213** and **214** with the space part **215** therebetween.
(362) In the embodiment illustrated in FIG. **9**, the magnet unit **240** is located adjacent to the fourth surface **214**. At this point, the first and second Halbach arrays **220** and **230** are located to be biased to the third surface **213**.
(363) In the embodiment illustrated in FIG. **10**, the magnet unit **240** is located adjacent to the third surface **213**. At this point, the first and second Halbach arrays **220** and **230** are located to be biased to the fourth surface **214**.
(364) The magnet unit **240** is disposed to face the any one surface of the third and fourth surfaces **213** and **214**. The space part **215**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **215** are located between the magnet unit **240** and the any one surface.
(365) The magnet unit **240** is located between the first surface **211** and the second surface **212**. In one embodiment, the magnet unit **240** may be located at a central portion of the other surface of the third and fourth surfaces **213** and **214**.
(366) In other words, the shortest distance between the magnet unit **240** and the first surface **211** and the shortest distance between the magnet unit **240** and the second surface **212** may be the same.
(367) The magnet unit **240** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and second Halbach arrays **220** and **230**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the magnet unit **240** is well known in the art, a detailed description thereof will be omitted.
(368) The magnet unit **240** is formed to extend in one direction. In the illustrated embodiment, the magnet unit **240** is formed to extend in a direction in which the third surface **213** or the fourth surface **214** extends, that is, in the front-rear direction.
(369) The magnet unit **240** includes a plurality of surfaces.
(370) Specifically, the magnet unit **240** includes the facing surface **241** facing the space part **215** or the fixed contactor **22** and the opposing surface **242** opposite to the space part **215** or the fixed contactor **22**.
(371) Each surface of the magnet unit **240** may be magnetized according to a predetermined rule.
(372) Specifically, the facing surface **241** may be magnetized to the same polarity as the first to

third inner surfaces **221**a, **222**a, and **223**a of the first Halbach array **220**. In addition, the facing surface **241** may be magnetized to the same polarity as the first to third inner surfaces **231**a, **232**a, and **233**a of the second Halbach array **230**.
(373) Similarly, the opposing surface **242** may be magnetized to the same polarity as the first to third outer surfaces **221**b, **222**b, and **223**b of the first Halbach array **220**. In addition, the opposing surface **242** may be magnetized to the same polarity as the first to third outer surfaces **231**b, **232**b, and **233**b of the second Halbach array **230**.
(374) At this point, it will be understood that the polarity of the facing surface **241** and the polarity of the opposing surface **242** are formed to be different from each other.
(375) Hereinafter, the arc path A.P formed by the arc path formation unit **200** according to the present embodiment will be described in detail with reference to FIGS. **11** and **12**.
(376) Referring to FIGS. **11** and **12**, the first to third inner surfaces **221**a, **222**a, and **223**a of the first Halbach array **220** are magnetized to N poles. According to the above-described rule, the first to third inner surfaces **231**a, **232**a, and **233**a of the second Halbach array **230** and the facing surface **241** of the magnet unit **240** are also magnetized to N poles.
(377) Accordingly, magnetic fields that repel each other are formed between the first Halbach array **220** and the second Halbach array **230**. In addition, in the magnet unit **240**, a magnetic field diverging in a direction from the facing surface **241** toward the space part **215** is formed.
(378) Accordingly, in the embodiment illustrated in FIG. **11**, the magnetic fields formed by the first Halbach array **220**, the second Halbach array **230**, and the magnet unit **240** are formed toward the third surface **213**, i.e., toward the left side in the illustrated embodiment.
(379) In addition, in the embodiment illustrated in FIG. **12**, the magnetic fields formed by the first Halbach array **220**, the second Halbach array **230**, and the magnet unit **240** are formed toward the fourth surface **214**, i.e., toward the right side in the illustrated embodiment.
(380) In the embodiment illustrated in FIG. **11**A, a direction of current is a direction from the second fixed contactor **22**b to the first fixed contactor **22**a via the movable contactor **43**.
(381) When the Fleming's left-hand rule is applied to the first fixed contactor **22**a, an electromagnetic force generated in the vicinity of the first fixed contactor **22**a is formed toward the front left side.
(382) Accordingly, an arc path A.P generated in the vicinity of the first fixed contactor **22**a is also formed toward the front left side.
(383) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22**b, an electromagnetic force generated in the vicinity of the second fixed contactor **22**b is formed toward the rear right side.
(384) Accordingly, an arc path A.P generated in the vicinity of the second fixed contactor **22**b is also formed toward the rear right side.
(385) In the embodiment illustrated in FIG. **11**B, a direction of current is a direction from the first fixed contactor **22**a to the second fixed contactor **22**b via the movable contactor **43**.
(386) When the Fleming's left-hand rule is applied to the first fixed contactor **22**a, an electromagnetic force generated in the vicinity of the first fixed contactor **22**a is formed toward the rear left side.
(387) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22**a is also formed toward the rear left side.
(388) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22**b, an electromagnetic force generated in the vicinity of the second fixed contactor **22**b is formed toward the front right side.
(389) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22**b is also formed toward the front right side.
(390) In the embodiment illustrated in FIG. **12**A, a direction of current is a direction from the second fixed contactor **22**b to the first fixed contactor **22**a via the movable contactor **43**.

(391) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(392) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(393) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(394) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(395) In the embodiment illustrated in FIG. **12**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(396) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(397) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(398) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(399) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(400) Although not illustrated in the drawing, when the polarity of each surface of the first Halbach array **220**, the second Halbach array **230**, and the magnet unit **240** is changed, the direction of the magnetic field formed in each of the first Halbach array **220**, the second Halbach array **230**, and the magnet unit **240** is reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(401) That is, in the electric connection situation shown in FIG. **11**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(402) Similarly, in the electric connection situation shown in FIG. **111**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(403) In addition, in the electric connection situation shown in FIG. **12**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(404) Similarly, in the electric connection situation shown in FIG. **12**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(405) Accordingly, in the arc path formation unit **200** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the first Halbach array **220**, the second Halbach array **230**, and the magnet unit **240** or the direction of the current flowing through the direct current relay **1**.
(406) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.

(3) Description of Arc Path Formation Unit **300** According Still Another Embodiment of Present Disclosure

(407) Hereinafter, an arc path formation unit **300** according to still another embodiment of the present disclosure will be described in detail with reference to FIGS. **13** to **20**.

(408) Referring to FIGS. **13** to **16**, the arc path formation unit **300** according to the illustrated embodiment includes a magnet frame **310**, a Halbach array **320**, a first magnet unit **330**, and a second magnet unit **340**.

(409) The magnet frame **310** according to the present embodiment has the same structure and function as the magnet frame **110** according to the above-described embodiment. However, there is a difference in the arrangement method of the Halbach array **320**, the first magnet unit **330**, and the second magnet unit **340** disposed in the magnet frame **310** according to the present embodiment.

(410) Accordingly, a description of the magnet frame **310** will be replaced with the description of the magnet frame **110** according to the above-described embodiment.

(411) In the illustrated embodiment, the plurality of magnetic materials constituting the Halbach array **320** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the Halbach array **320** is formed to extend in the left-right direction.

(412) The Halbach array **320** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the Halbach array **320** may form a magnetic field together with the first and second magnet units **330** and **340**.

(413) The Halbach array **320** may be located adjacent to any one surface of first and second surfaces **311** and **312**. In one embodiment, the Halbach array **320** may be coupled to an inner side (i.e., the side in a direction toward a space part **315**) of the any one surface.

(414) In the embodiment illustrated in FIGS. **13** and **15**, the Halbach array **320** is disposed on an inner side of the second surface **312** and adjacent to the second surface **312** and faces the first magnet unit **330** located on an inner side of the first surface **311**.

(415) In the embodiment illustrated in FIGS. **14** and **16**, the Halbach array **320** is disposed on the inner side of the first surface **311** and adjacent to the first surface **311** and faces the first magnet unit **330** located on the inner side of the second surface **312**.

(416) The space part **315**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **315** are located between the Halbach array **320** and the first magnet unit **330**.

(417) In addition, the space part **315**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **315** are located between the Halbach array **320** and the second magnet unit **340**.

(418) The Halbach array **320** may be located to be biased to any one surface of a third surface **313** and a fourth surface **314**. In the embodiment illustrated in FIGS. **13** and **14**, the Halbach array **320** is located to be biased to the third surface **313**. In the embodiment illustrated in FIGS. **15** and **16**, the Halbach array **320** is located to be biased to the fourth surface **314**.

(419) The Halbach array **320** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and second magnet units **330** and **340**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the Halbach array **320** is a well-known technique, a detailed description thereof will be omitted.

(420) In the illustrated embodiment, the first Halbach array **320** includes a first block **321**, a second block **322**, and a third block **323**. It will be understood that the plurality of magnetic materials constituting the Halbach array **320** are named as the blocks **321**, **322**, and **323**, respectively.

(421) The first to third blocks as the blocks **321**, **322**, and **323** may each be formed of a magnetic material. In one embodiment, the first to third blocks as the blocks **321**, **322**, and **323** may each be provided as a permanent magnet, an electromagnet, or the like.

(422) The first to third blocks as the blocks **321**, **322**, and **323** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks as the blocks **321**, **322**, and **323** are disposed in parallel in a direction in which the first surface **311** or the second surface **312** extends,

that is, in the left-right direction.
(423) The first block **321** is located adjacent to the any one surface of the third surface **313** and the fourth surface **314**. In addition, the third block **323** is located adjacent to the other surface of the third surface **313** and the fourth surface **314**. The second block **322** is located between the first block **321** and the third block **323**.
(424) In one embodiment, the blocks **321**, **322**, and **323** adjacent to each other may be in contact with each other.
(425) The second block **322** may be disposed to overlap the first magnet unit **330** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the first magnet unit **330** or the space part **315**, i.e., in the front-rear direction in the illustrated embodiment.
(426) Each of the blocks **321**, **322**, and **323** includes a plurality of surfaces.
(427) Specifically, the first block **321** includes a first inner surface **321***a* facing the second block **322** and a first outer surface **321***b* opposite to the second block **322**.
(428) The second block **322** includes a second inner surface **322***a* facing the space part **315** or the first magnet unit **330** and a second outer surface **322***b* opposite to the space part **315** or the first magnet unit **330**.
(429) The third block **323** includes a third inner surface **323***a* facing the second block **322** and a third outer surface **323***b* opposite to the second block **322**.
(430) The plurality of surfaces of each of the blocks **321**, **322**, and **323** may be magnetized according to a predetermined rule to configure a Halbach array.
(431) Specifically, the first to third inner surfaces **321***a*, **322***a*, and **323***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **321***b*, **322***b*, and **323***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **321***a*, **322***a*, and **323***a*.
(432) At this point, the first to third inner surfaces **321***a*, **322***a*, and **323***a* may be magnetized to the same polarity as a first facing surface **331** of the first magnet unit **330** and a second facing surface **341** of the second magnet unit **340**.
(433) Similarly, the first to third outer surfaces **321***b*, **322***b*, and **323***b* may be magnetized to the same polarity as a first opposing surface **332** of the first magnet unit **330** and a second opposing surface **342** of the second magnet unit **340**.
(434) The first magnet unit **330** forms a magnetic field by itself, or forms magnetic fields together with the Halbach array **320** and the second magnet unit **340**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the first magnet unit **330**.
(435) The first magnet unit **330** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the first magnet unit **330** may be provided as a permanent magnet, an electromagnet, or the like.
(436) The first magnet unit **330** may be located adjacent to the other surface of the first and second surfaces **311** and **312**. In one embodiment, the first magnet unit **330** may be coupled to an inner side (i.e., the side in a direction toward the space part **315**) of the other surface.
(437) At this point, the first magnet unit **330** is located on a surface opposite to any one surface of the first and second surfaces **311** and **312**, to which the Halbach array **320** is located adjacent.
(438) In the embodiment illustrated in FIGS. **13** and **15**, the first magnet unit **330** is located adjacent to the first surface **311** and is disposed to face the Halbach array **320** located adjacent to the second surface **312**.
(439) In the embodiment illustrated in FIGS. **14** and **16**, the first magnet unit **330** is located adjacent to the second surface **312** and is disposed to face the Halbach array **320** located adjacent to the first surface **311**.
(440) The space part **315**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **315** are located between the first magnet unit **330** and the Halbach array **320**.
(441) In addition, the space part **315**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **315** are located between the first magnet unit **330** and the second

magnet unit **340**.
(442) The first magnet unit **330** may be located to be biased to the any one surface of the third surface **313** and the fourth surface **314**. In the embodiment illustrated in FIGS. **13** and **14**, the first magnet unit **330** is located to be biased to the third surface **313**. In the embodiment illustrated in FIGS. **15** and **16**, the first magnet unit **330** is located to be biased to the fourth surface **314**.
(443) The first magnet unit **330** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the Halbach array **320** and the second magnet unit **340**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first magnet unit **330** is well known in the art, a detailed description thereof will be omitted.
(444) The first magnet unit **330** is formed to extend in one direction. In the illustrated embodiment, the first magnet unit **330** is formed to extend in a direction in which the first surface **311** or the second surface **312** extends, that is, in the left-right direction.
(445) The first magnet unit **330** includes a plurality of surfaces.
(446) Specifically, the first magnet unit **330** includes the first facing surface **331** facing the space part **315** or the fixed contactor **22** and the first opposing surface **332** opposite to the space part **315** or the fixed contactor **22**.
(447) Each surface of the first magnet unit **330** may be magnetized according to a predetermined rule.
(448) Specifically, the first facing surface **331** may be magnetized to the same polarity as the first to third inner surfaces **321***a*, **322***a*, and **323***a* of the Halbach array **320**. In addition, the first facing surface **331** may be magnetized to the same polarity as the second facing surface **341** of the second magnet unit **340**.
(449) Similarly, the first opposing surface **332** may be magnetized to the same polarity as the first to third outer surfaces **321***b*, **322***b*, and **323***b* of the Halbach array **320**. In addition, the first opposing surface **332** may be magnetized to the same polarity as the second opposing surface **342** of the second magnet unit **340**.
(450) At this point, it will be understood that the polarity of the second facing surface **341** and the polarity of the second opposing surface **342** are formed to be different from each other.
(451) The second magnet unit **340** forms a magnetic field by itself, or forms magnetic fields together with the Halbach array **320** and the first magnet unit **330**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the second magnet unit **340**.
(452) The second magnet unit **340** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the second magnet unit **340** may be provided as a permanent magnet, an electromagnet, or the like.
(453) The second magnet unit **340** may be located adjacent to the other surface of the third and fourth surfaces **313** and **314**. In one embodiment, the second magnet unit **340** may be coupled to an inner side (i.e., the side in a direction toward the space part **315**) of the other surface.
(454) At this point, the second magnet unit **340** is located on a surface opposite to the any one surface of the third and fourth surfaces **313** and **314**, to which the Halbach array **320** and the first magnet unit **330** are located to be biased.
(455) The second magnet unit **340** is disposed to face the other surface of the third and fourth surfaces **313** and **314** with the space part **315** therebetween.
(456) In the embodiment illustrated in FIGS. **13** and **14**, the second magnet unit **340** is located adjacent to the fourth surface **314**. At this point, the Halbach array **320** and the first magnet unit **330** are located to be biased to the third surface **313**.
(457) In the embodiment illustrated in FIGS. **15** and **16**, the second magnet unit **340** is located adjacent to the third surface **313**. At this point, the Halbach array **320** and the first magnet unit **330** are located to be biased to the fourth surface **314**.
(458) The second magnet unit **340** is disposed to face the other surface of the third and fourth

surfaces **313** and **314**. The space part **315**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **315** are located between the second magnet unit **340** and the other surface.
(459) The second magnet unit **340** is located between the first surface **311** and the second surface **312**. In one embodiment, the second magnet unit **340** may be located at a central portion of the other surface of the third and fourth surfaces **313** and **314**.
(460) In other words, the shortest distance between the second magnet unit **340** and the first surface **311** and the shortest distance between the second magnet unit **340** and the second surface **312** may be the same.
(461) The second magnet unit **340** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the Halbach array **320** and the first magnet unit **330**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second magnet unit **340** is well known in the art, a detailed description thereof will be omitted.
(462) The second magnet unit **340** is formed to extend in the other direction. In the illustrated embodiment, the second magnet unit **340** is formed to extend in a direction in which the third surface **313** or the fourth surface **314** extends, that is, in the front-rear direction.
(463) The second magnet unit **340** includes a plurality of surfaces.
(464) Specifically, the second magnet unit **340** includes the second facing surface **341** facing the space part **315** or the fixed contactor **22** and the second opposing surface **342** opposite to the space part **315** or the fixed contactor **22**.
(465) Each surface of the second magnet unit **340** may be magnetized according to a predetermined rule.
(466) Specifically, the second facing surface **341** may be magnetized to the same polarity as the first to third inner surfaces **321***a*, **322***a*, and **323***a* of the Halbach array **320**. In addition, the second facing surface **341** may be magnetized to the same polarity as the first facing surface **331** of the first magnet unit **330**.
(467) Similarly, the second opposing surface **342** may be magnetized to the same polarity as the first to third outer surfaces **321***b*, **322***b*, and **323***b* of the Halbach array **320**. In addition, the second opposing surface **342** may be magnetized to the same polarity as the first opposing surface **332** of the first magnet unit **330**.
(468) At this point, it will be understood that the polarity of the second facing surface **341** and the polarity of the second opposing surface **342** are formed to be different from each other.
(469) Hereinafter, an arc path A.P formed by the arc path formation unit **300** according to the present embodiment will be described in detail with reference to FIGS. **17** to **20**.
(470) Referring to FIGS. **17** to **20**, the first to third inner surfaces **321***a*, **322***a*, and **323***a* of the Halbach array **320** are magnetized to N poles. According to the above-described rule, the first facing surface **331** of the first magnet unit **330** and the second facing surface **341** of the second magnet unit **340** are also magnetized to N poles.
(471) Accordingly, magnetic fields that repel each other are formed between the Halbach array **320** and the first magnet unit **330**. In addition, in the second magnet unit **340**, a magnetic field diverging in a direction from the second facing surface **341** toward the space part **315** is formed.
(472) Accordingly, in the embodiment illustrated in FIGS. **17** and **18**, the magnetic fields formed by the Halbach array **320** and the first and second magnet units **330** and **340** are formed toward the third surface **313**, i.e., toward the left side in the illustrated embodiment.
(473) In addition, in the embodiment illustrated in FIGS. **19** and **20**, the magnetic fields formed by the Halbach array **320** and the first and second magnet units **330** and **340** are formed toward the fourth surface **314**, i.e., toward the right side in the illustrated embodiment.
(474) In the embodiment illustrated in FIGS. **17**A and **18**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.

(475) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(476) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(477) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(478) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(479) In addition, in the embodiment illustrated in FIGS. **19**A and **20**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(480) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(481) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(482) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(483) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(484) In the embodiment illustrated in FIGS. **17**B and **18**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(485) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(486) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(487) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(488) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(489) In the embodiment illustrated in FIGS. **19**B and **20**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(490) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(491) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(492) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(493) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(494) Although not illustrated in the drawing, when the polarity of each surface of the Halbach array **320** and the first and second magnet units **330** and **340** is changed, the direction of the magnetic field formed in each of the Halbach array **320** and the first and second magnet units **330**

and **340** is reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(495) That is, in the electric connection situation shown in FIGS. **17**A and **18**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(496) Similarly, in the electric connection situation shown in FIGS. **17**B and **18**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(497) In addition, in the electric connection situation shown in FIGS. **19**A and **20**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(498) Similarly, in the electric connection situation shown in FIGS. **19**B and **20**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(499) Accordingly, in the arc path formation unit **300** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the Halbach array **320** and the first and second magnet units **330** and **340** or the direction of the current flowing through the direct current relay **1**.
(500) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.
(4) Description of Arc Path Formation Unit **400** According to Yet Another Embodiment of Present Disclosure
(501) Hereinafter, an arc path formation unit **400** according to yet another embodiment of the present disclosure will be described in detail with reference to FIGS. **21** to **24**.
(502) Referring to FIGS. **21** and **22**, the arc path formation unit **400** according to the illustrated embodiment includes a magnet frame **410**, a Halbach array **420**, a first magnet unit **430**, and a second magnet unit **440**.
(503) The magnet frame **410** according to the present embodiment has the same structure and function as the magnet frame **110** according to the above-described embodiment. However, there is a difference in the arrangement method of the Halbach array **420**, the first magnet unit **430**, and the second magnet unit **440** disposed in the magnet frame **410** according to the present embodiment.
(504) Accordingly, a description of the magnet frame **410** will be replaced with the description of the magnet frame **110** according to the above-described embodiment.
(505) In the illustrated embodiment, the plurality of magnetic materials constituting the Halbach array **420** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the Halbach array **420** is formed to extend in the left-right direction.
(506) The Halbach array **420** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the Halbach array **420** may form magnetic fields together with the first and second magnet units **430** and **440**.
(507) The Halbach array **420** may be located adjacent to any one surface of third and fourth surfaces **413** and **414**. In one embodiment, the Halbach array **420** may be coupled to an inner side (i.e., the side in a direction toward a space part **415**) of the any one surface.
(508) At this point, the Halbach array **420** is located on a surface opposite to the other surface of the third and fourth surfaces **413** and **414**, to which first and second magnet units **430** and **440** are located to be biased.

(509) The Halbach array **420** is disposed to face the other surface of the third and fourth surfaces **413** and **414** with the space part **415** therebetween.
(510) In the embodiment illustrated in FIG. **21**, Halbach array **420** is located adjacent to the fourth surface **414**. At this point, the first and second magnet units **430** and **440** are located to be biased to the third surface **413**.
(511) In the embodiment illustrated in FIG. **22**, the Halbach array **420** is located adjacent to the third surface **413**. At this point, the first and second magnet units **430** and **440** are located to be biased to the fourth surface **414**.
(512) The Halbach array **420** is disposed to face any one surface of the third and fourth surfaces **413** and **414**. The space part **415**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **415** are located between the Halbach array **420** and the any one surface.
(513) The Halbach array **420** is located between a first surface **411** and a second surface **412**. In one embodiment, the Halbach array **420** may be located at a central portion of the other surface of the third and fourth surfaces **413** and **414**.
(514) In other words, the shortest distance between the Halbach array **420** and the first surface **411** and the shortest distance between the Halbach array **420** and the second surface **412** may be the same.
(515) In the illustrated embodiment, the Halbach array **420** includes a first block **421**, a second block **422**, and a third block **423**. It will be understood that the plurality of magnetic materials constituting the Halbach array **420** are named as the blocks **421**, **422**, and **423**, respectively.
(516) The first to third blocks **421**, **422**, and **423** may each be formed of a magnetic material. In one embodiment, the first to third blocks **421**, **422**, and **423** may each be provided as a permanent magnet, an electromagnet, or the like.
(517) The first to third blocks **421**, **422**, and **423** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **421**, **422**, and **423** are disposed in parallel in a direction in which the third surface **413** or the fourth surface **414** extends, that is, in the left-right direction.
(518) The first block **421** is located on the rear side. That is, the first block **421** is located adjacent to the first surface **411**. In addition, the third block **423** is located at the most front side. That is, the third block **423** is located adjacent to the second surface **412**. The second block **422** is located between the first block **421** and the third block **423**.
(519) In one embodiment, the blocks **421**, **422**, and **423** adjacent to each other may be in contact with each other.
(520) The second block **422** may be disposed to overlap the fixed contactors **22***a* and **22***b* in a direction toward the space part **415**, i.e., in the left-right direction in the illustrated embodiment.
(521) Each of the blocks **421**, **422**, and **423** includes a plurality of surfaces.
(522) Specifically, the first block **421** includes a first inner surface **421***a* facing the second block **422** and a first outer surface **421***b* opposite to the second block **422**.
(523) The second block **422** includes a second inner surface **422***a* facing the space part **415** and a second outer surface **422***b* opposite to the space part **415**.
(524) The third block **423** includes a third inner surface **423***a* facing the second block **422** and a third outer surface **423***b* opposite to the second block **422**.
(525) The plurality of surfaces of each of the blocks **421**, **422**, and **423** may be magnetized according to a predetermined rule to configure a Halbach array.
(526) Specifically, the first to third inner surfaces **421***a*, **422***a*, and **423***a* may be magnetized to the same polarity. In addition, the first to third outer surfaces **421***b*, **422***b*, and **423***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **421***a*, **422***a*, and **423***a*.
(527) At this point, the first to third inner surfaces **421***a*, **422***a*, and **423***a* may be magnetized to the same polarity as a first facing surface **431** of the first magnet unit **430** and a second facing surface

441 of the second magnet unit **440**.
(528) Similarly, the first to third outer surfaces **421***b*, **422***b*, and **423***b* may be magnetized to the same polarity as a first opposing surface **432** of the first magnet unit **430** and a second opposing surface **442** of the second magnet unit **440**.
(529) The first magnet unit **430** forms a magnetic field by itself, or forms magnetic fields together with the Halbach array **420** and the second magnet unit **440**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the first magnet unit **430**.
(530) The first magnet unit **430** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the first magnet unit **430** may be provided as a permanent magnet, an electromagnet, or the like.
(531) The first magnet unit **430** may be located adjacent to any one surface of the first and second surfaces **411** and **412**. In one embodiment, the first magnet unit **430** may be coupled to an inner side (i.e., the side in a direction toward the space part **415**) of the any one surface.
(532) At this point, the first magnet unit **430** is located on a surface opposite to any one surface of the first and second surfaces **411** and **412**, to which the second magnet unit **440** is located adjacent.
(533) In the embodiment illustrated in FIG. **21**, the first magnet unit **430** is located adjacent to the first surface **411** and is disposed to face the second magnet unit **440** located adjacent to the second surface **412**.
(534) In the embodiment illustrated in FIG. **22**, the first magnet unit **430** is located adjacent to the second surface **412** and is disposed to face the second magnet unit **440** located adjacent to the first surface **411**.
(535) The space part **415**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **415** are located between the first magnet unit **430** and the second magnet unit **440**.
(536) In addition, the space part **415**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **415** are located between the first magnet unit **430** and the Halbach array **420**.
(537) The first magnet unit **430** may be located to be biased to the other surface of the third surface **413** and the fourth surface **414**.
(538) In the embodiment illustrated in FIG. **21**, the first magnet unit **430** is located to be biased to the third surface **413**. In the embodiment illustrated in FIG. **22**, the first magnet unit **430** is located to be biased to the fourth surface **414**.
(539) The first magnet unit **430** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the Halbach array **420** and the second magnet unit **440**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first magnet unit **430** is well known in the art, a detailed description thereof will be omitted.
(540) The first magnet unit **430** is formed to extend in one direction. In the illustrated embodiment, the first magnet unit **430** is formed to extend in a direction in which the first surface **411** or the second surface **412** extends, that is, in the left-right direction.
(541) The first magnet unit **430** includes a plurality of surfaces.
(542) Specifically, the first magnet unit **430** includes the first facing surface **431** facing the space part **415** or the fixed contactor **22** and the first opposing surface **432** opposite to the space part **415** or the fixed contactor **22**.
(543) Each surface of the first magnet unit **430** may be magnetized according to a predetermined rule.
(544) Specifically, the first facing surface **431** may be magnetized to the same polarity as the first to third inner surfaces **421***a*, **422***a*, and **423***a* of the Halbach array **420**. In addition, the first facing surface **431** may be magnetized to the same polarity as the second facing surface **441** of the second magnet unit **440**.
(545) Similarly, the first opposing surface **432** may be magnetized to the same polarity as the first

to third outer surfaces **421***b*, **422***b*, and **423***b* of the Halbach array **420**. In addition, the first opposing surface **432** may be magnetized to the same polarity as the second opposing surface **442** of the second magnet unit **440**.

(546) At this point, it will be understood that the polarity of the first facing surface **431** and the polarity of the first opposing surface **432** are formed to be different from each other.

(547) The second magnet unit **440** forms a magnetic field by itself, or forms magnetic fields together with the Halbach array **420** and the first magnet unit **430**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the second magnet unit **440**.

(548) The second magnet unit **440** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the second magnet unit **440** may be provided as a permanent magnet, an electromagnet, or the like.

(549) The second magnet unit **440** may be located adjacent to the other surface of the first and second surfaces **411** and **412**. In one embodiment, the second magnet unit **440** may be coupled to an inner side (i.e., the side in a direction toward the space part **415**) of the other surface.

(550) At this point, the second magnet unit **440** is located on a surface opposite to any one surface of the first and second surfaces **411** and **412**, to which the first magnet unit **430** is located adjacent.

(551) In the embodiment illustrated in FIG. **21**, the second magnet unit **440** is located adjacent to the first surface **411** and is disposed to face the first magnet unit **430** located adjacent to the second surface **412**.

(552) In the embodiment illustrated in FIG. **22**, the second magnet unit **440** is located adjacent to the second surface **412** and is disposed to face the first magnet unit **430** located adjacent to the first surface **411**.

(553) The space part **415**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **415** are located between the second magnet unit **440** and the first magnet unit **430**.

(554) In addition, the space part **415**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **415** are located between the second magnet unit **440** and the Halbach array **420**.

(555) The second magnet unit **440** may be located to be biased to the other surface of the third surface **413** and the fourth surface **414**.

(556) In the embodiment illustrated in FIG. **21**, the second magnet unit **440** is located to be biased to the third surface **413**. In the embodiment illustrated in FIG. **22**, the second magnet unit **440** is located to be biased to the fourth surface **414**.

(557) The second magnet unit **440** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the Halbach array **420** and the first magnet unit **430**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second magnet unit **440** is well known in the art, a detailed description thereof will be omitted.

(558) The second magnet unit **440** is formed to extend in one direction. In the illustrated embodiment, the second magnet unit **440** is formed to extend in a direction in which the first surface **411** or the second surface **412** extends, that is, in the left-right direction.

(559) The second magnet unit **440** includes a plurality of surfaces.

(560) Specifically, the second magnet unit **440** includes the second facing surface **441** facing the space part **415** or the fixed contactor **22** and the second opposing surface **442** opposite to the space part **415** or the fixed contactor **22**.

(561) Each surface of the second magnet unit **440** may be magnetized according to a predetermined rule.

(562) Specifically, the second facing surface **441** may be magnetized to the same polarity as the first to third inner surfaces **421***a*, **422***a*, and **423***a* of the Halbach array **420**. In addition, the second facing surface **441** may be magnetized to the same polarity as the first facing surface **431** of the first magnet unit **430**.

(563) Similarly, the second opposing surface **442** may be magnetized to the same polarity as the first to third outer surfaces **421***b*, **422***b*, and **423***b* of the Halbach array **420**. In addition, the second opposing surface **442** may be magnetized to the same polarity as the first opposing surface **432** of the first magnet unit **430**.
(564) At this point, it will be understood that the polarity of the second facing surface **441** and the polarity of the second opposing surface **442** are formed to be different from each other.
(565) Hereinafter, an arc path A.P formed by the arc path formation unit **400** according to the present embodiment will be described in detail with reference to FIGS. **23** and **24**.
(566) Referring to FIGS. **23** and **24**, the first to third inner surfaces **421***a*, **422***a*, and **423***a* of the Halbach array **420** are magnetized to N poles. According to the above-described rule, the first facing surface **431** of the first magnet unit **430** and the second facing surface **441** of the second magnet unit **440** are also magnetized to N poles.
(567) Accordingly, magnetic fields that repel each other are formed between the first magnet unit **430** and the second magnet unit **440**. In addition, in the Halbach array **420**, a magnetic field diverging in a direction from the second inner surface **422***a* is formed.
(568) Accordingly, in the embodiment illustrated in FIG. **23**, the magnetic fields formed by the Halbach array **420** and the first and second magnet units **430** and **440** are formed toward the third surface **413**, i.e., toward the left side in the illustrated embodiment.
(569) In addition, in the embodiment illustrated in FIG. **24**, the magnetic fields formed by the Halbach array **420** and the first and second magnet units **430** and **440** are formed toward the fourth surface **414**, i.e., toward the right side in the illustrated embodiment.
(570) In the embodiment illustrated in FIG. **23**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(571) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(572) Accordingly, an arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(573) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(574) Accordingly, an arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(575) In the embodiment illustrated in FIG. **23**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(576) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(577) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(578) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(579) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(580) In addition, in the embodiment illustrated in FIG. **24**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(581) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.

(582) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(583) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(584) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(585) In the embodiment illustrated in FIG. **24**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(586) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(587) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(588) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(589) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(590) Although not illustrated in the drawing, when the polarity of each surface of the Halbach array **420** and the first and second magnet units **430** and **440** is changed, the direction of the magnetic field formed in each of the Halbach array **420** and the first and second magnet units **430** and **440** is reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(591) That is, in the electric connection situation shown in FIG. **23**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(592) Similarly, in the electric connection situation shown in FIG. **23**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(593) In addition, in the electric connection situation shown in FIG. **24**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(594) Similarly, in the electric connection situation shown in FIG. **24**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(595) Accordingly, in the arc path formation unit **400** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the Halbach array **420** and the first and second magnet units **430** and **440** or the direction of the current flowing through the direct current relay **1**.
(596) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.
(5) Description of Arc Path Formation Unit **500** According to Yet Another Embodiment of Present Disclosure
(597) Hereinafter, an arc path formation unit **500** according to yet another embodiment of the

present disclosure will be described in detail with reference to FIGS. **25** to **28**.
(598) Referring to FIGS. **25** and **26**, the arc path formation unit **500** according to the illustrated embodiment includes a magnet frame **510**, a first Halbach array **520**, a second Halbach array **530**, a third Halbach array **540**, and a magnet unit **550**.
(599) The magnet frame **510** according to the present embodiment has the same structure and function as the magnet frame **110** according to the above-described embodiment. However, there is a difference in the arrangement method of the first Halbach array **520**, the second Halbach array **530**, the third Halbach array **540**, and the magnet unit **550** disposed in the magnet frame **510** according to the present embodiment.
(600) Accordingly, a description of the magnet frame **510** will be replaced with the description of the magnet frame **110** according to the above-described embodiment.
(601) In the illustrated embodiment, the plurality of magnetic materials constituting the first Halbach array **520** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the first Halbach array **520** is formed to extend in the left-right direction.
(602) The first Halbach array **520** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the first Halbach array **520** may form magnetic fields together with the second and third Halbach arrays **530** and **540** and the magnet unit **550**.
(603) The first Halbach array **520** may be located adjacent to any one surface of first and second surfaces **511** and **512**. In one embodiment, the first Halbach array **520** may be coupled to an inner side (i.e., the side in a direction toward a space part **515**) of the any one surface.
(604) In the illustrated embodiment, the first Halbach array **520** is disposed on an inner side of the first surface **511** and adjacent to the first surface **511** and faces the second Halbach array **530** located on an inner side of the second surface **512**.
(605) The space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the first Halbach array **520** and the second Halbach array **530**.
(606) In addition, the space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the first Halbach array **520** and the third Halbach array **540**.
(607) Furthermore, the space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the first Halbach array **520** and the magnet unit **550**.
(608) The first Halbach array **520** may be located to be biased to any one surface of a third surface **513** and a fourth surface **514**. In the embodiment illustrated in FIG. **25**, the first Halbach array **520** is located to be biased to the third surface **513**. In the embodiment illustrated in FIG. **26**, the first Halbach array **520** is located to be biased to the fourth surface **514**.
(609) The any one surface, to which the first Halbach array **520** is located to be biased, may be a surface to which the magnet unit **550** is located adjacent. In addition, the any one surface, to which the first Halbach array **520** is located to be biased, may be a surface opposite to a surface to which the third Halbach array **540** is located adjacent.
(610) The first Halbach array **520** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the second and third Halbach arrays **530** and **540** and the magnet unit **550**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first Halbach array **520** is a well-known technique, a detailed description thereof will be omitted.
(611) In the illustrated embodiment, the first Halbach array **520** includes a first block **521**, a second block **522**, and a third block **523**. It will be understood that the plurality of magnetic materials constituting the first Halbach array **520** are named as the blocks **521**, **522**, and **523**, respectively.
(612) The first to third blocks **521**, **522**, and **523** may each be formed of a magnetic material. In one

embodiment, the first to third blocks **521**, **522**, and **523** may each be provided as a permanent magnet, an electromagnet, or the like.
(613) The first to third blocks **521**, **522**, and **523** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **521**, **522**, and **523** are disposed in parallel in a direction in which the first surface **511** extends, that is, in the left-right direction.
(614) The first block **521** is located adjacent to the any one surface of the third surface **513** and the fourth surface **514**. In addition, the third block **523** is located adjacent to the other surface of the third surface **513** and the fourth surface **514**. The second block **522** is located between the first block **521** and the third block **523**.
(615) In one embodiment, the blocks **521**, **522**, and **523** adjacent to each other may be in contact with each other.
(616) The second block **522** may be disposed to overlap a second block **532** of the second Halbach array **530** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the second Halbach array **530** or the space part **515**, i.e., in the front-rear direction in the illustrated embodiment.
(617) Each of the blocks **521**, **522**, and **523** includes a plurality of surfaces.
(618) Specifically, the first block **521** includes a first inner surface **521***a* facing the second block **522** and a first outer surface **521***b* opposite to the second block **522**.
(619) The second block **522** includes a second inner surface **522***a* facing the space part **515** or the second Halbach array **530** and a second outer surface **522***b* opposite to the space part **515** or the second Halbach array **530**.
(620) The third block **523** includes a third inner surface **523***a* facing the second block **522** and a third outer surface **523***b* opposite to the second block **522**.
(621) The plurality of surfaces of each of the blocks **521**, **522**, and **523** may be magnetized according to a predetermined rule to configure a Halbach array.
(622) Specifically, the first to third inner surfaces **521***a*, **522***a*, and **523***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **521***b*, **522***b*, and **523***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **521***a*, **522***a*, and **523***a*.
(623) At this point, the first to third inner surfaces **521***a*, **522***a*, and **523***a* may be magnetized to the same polarity as first to third inner surfaces **531***a*, **532***a*, and **533***a* of the second Halbach array **530** and first to third inner surfaces **541***a*, **542***a*, and **543***a* of the third Halbach array **540**. In addition, the first to third inner surfaces **521***a*, **522***a*, and **523***a* may be magnetized to the same polarity as an opposing surface **552** of the magnet unit **550**.
(624) Similarly, the first to third outer surfaces **521***b*, **522***b*, and **523***b* may be magnetized to the same polarity as first to third outer surfaces **531***b*, **532***b*, and **533***b* of the second Halbach array **530** and first to third outer surfaces **541***b*, **542***b*, and **543***b* of the third Halbach array **540**. In addition, the first to third outer surfaces **541***b*, **542***b*, and **543***b* may be magnetized to the same polarity as a facing surface **551** of the magnet unit **550**.
(625) In the illustrated embodiment, the plurality of magnetic materials constituting the second Halbach array **530** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the second Halbach array **530** is formed to extend in the left-right direction.
(626) The second Halbach array **530** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the second Halbach array **530** may form magnetic fields together with the first and third Halbach arrays **520** and **540** and the magnet unit **550**.
(627) The second Halbach array **530** may be located adjacent to the other surface of the first and second surfaces **511** and **512**. In one embodiment, the second Halbach array **530** may be coupled to an inner side (i.e., the side in a direction toward the space part **515**) of the other surface.
(628) In the illustrated embodiment, the second Halbach array **530** is disposed on the inner side of the second surface **512** and adjacent to the second surface **512** and faces the first Halbach array **520** located on the inner side of the first surface **511**.

(629) The space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the second Halbach array **530** and the first Halbach array **520**.
(630) In addition, the space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the second Halbach array **530** and the third Halbach array **540**.
(631) Furthermore, the space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the second Halbach array **530** and the magnet unit **550**.
(632) The second Halbach array **530** may be located to be biased to any one surface of the third surface **513** and the fourth surface **514**. In the embodiment illustrated in FIG. **25**, the second Halbach array **530** is located to be biased to the third surface **513**. In the embodiment illustrated in FIG. **26**, the second Halbach array **530** is located to be biased to the fourth surface **514**.
(633) The any one surface, to which the second Halbach array **530** is located to be biased, may be a surface to which the magnet unit **550** is located adjacent. In addition, the any one surface, to which the second Halbach array **530** is located to be biased, may be a surface opposite to a surface to which the third Halbach array **540** is located adjacent.
(634) The second Halbach array **530** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and third Halbach arrays **520** and **540** and the magnet unit **550**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second Halbach array **530** is a well-known technique, a detailed description thereof will be omitted.
(635) In the illustrated embodiment, the second Halbach array **530** includes a first block **531**, the second block **532**, and a third block **533**. It will be understood that the plurality of magnetic materials constituting the second Halbach array **530** are named as the blocks **531**, **532**, and **533**, respectively.
(636) The first to third blocks **531**, **532**, and **533** may each be formed of a magnetic material. In one embodiment, the first to third blocks **531**, **532**, and **533** may each be provided as a permanent magnet, an electromagnet, or the like.
(637) The first to third blocks **531**, **532**, and **533** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **531**, **532**, and **533** are disposed in parallel in a direction in which the second surface **512** extends, that is, in the left-right direction.
(638) The first block **531** is located adjacent to the any one surface of the third surface **513** and the fourth surface **514**. In addition, the third block **533** is located adjacent to the other surface of the third surface **513** and the fourth surface **514**. The second block **532** is located between the first block **531** and the third block **533**.
(639) In one embodiment, the blocks **531**, **532**, and **533** adjacent to each other may be in contact with each other.
(640) The second block **532** may be disposed to overlap the second block **522** of the first Halbach array **520** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the first Halbach array **520** or the space part **515**, i.e., in the front-rear direction in the illustrated embodiment.
(641) Each of the blocks **531**, **532**, and **533** includes a plurality of surfaces.
(642) Specifically, the first block **531** includes the first inner surface **531***a* facing the second block **532** and the first outer surface **531***b* opposite to the second block **532**.
(643) The second block **532** includes the second inner surface **532***a* facing the space part **515** or the first Halbach array **520**, and the second outer surface **532***b* opposite to the space part **515** or the first Halbach array **520**.
(644) The third block **533** includes the third inner surface **533***a* facing the second block **532** and the third outer surface **533***b* opposite to the second block **532**.
(645) The plurality of surfaces of each of the blocks **531**, **532**, and **533** may be magnetized

according to a predetermined rule to configure a Halbach array.
(646) Specifically, the first to third inner surfaces **531***a*, **532***a*, and **533***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **531***b*, **532***b*, and **533***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **531***a*, **532***a*, and **533***a*.
(647) At this point, the first to third inner surfaces **531***a*, **532***a*, and **533***a* may be magnetized to the same polarity as the first to third inner surfaces **521***a*, **522***a*, and **523***a* of the first Halbach array **520** and the first to third inner surfaces **541***a*, **542***a*, and **543***a* of the third Halbach array **540**. In addition, the first to third inner surfaces **531***a*, **532***a*, and **533***a* may be magnetized to the same polarity as the opposing surface **552** of the magnet unit **550**.
(648) Similarly, the first to third outer surfaces **531***b*, **532***b*, and **533***b* may be magnetized to the same polarity as the first to third outer surfaces **521***b*, **522***b*, and **523***b* of the first Halbach array **520** and the first to third outer surfaces **541***b*, **542***b*, and **543***b* of the third Halbach array **540**. In addition, the first to third outer surfaces **541***b*, **542***b*, and **543***b* may be magnetized to the same polarity as the facing surface **551** of the magnet unit **550**.
(649) The third Halbach array **540** may be defined as an assembly of a plurality of magnetic materials. The plurality of magnetic materials included in the third Halbach array **540** may be disposed with predetermined regularity. In the illustrated embodiment, the plurality of magnetic materials constituting the third Halbach array **540** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the third Halbach array **540** is formed to extend in the left-right direction.
(650) The third Halbach array **540** may form a magnetic field by itself. That is, a magnetic field may be formed between the plurality of magnetic materials included in the third Halbach array **540**.
(651) In addition, the third Halbach array **540** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the third Halbach array **540** may form magnetic fields together with the first and second Halbach arrays **520** and **530** and the magnet unit **550**.
(652) The third Halbach array **540** may be located adjacent to the other surface of the third and fourth surfaces **513** and **514**. In one embodiment, the third Halbach array **540** may be coupled to an inner side (i.e., the side in a direction toward a space part **515**) of the any one surface.
(653) At this point, the third Halbach array **540** is located on a surface opposite to any one surface of the third and fourth surfaces **513** and **514**, to which the first and second Halbach arrays **520** and **530** are located to be biased.
(654) In the embodiment illustrated in FIG. **25**, the third Halbach array **540** is located adjacent to the fourth surface **514**. At this point, the first and second Halbach arrays **520** and **530** are located to be biased to the third surface **513**.
(655) In the embodiment illustrated in FIG. **26**, the third Halbach array **540** is located adjacent to the third surface **513**. At this point, the first and second Halbach arrays **520** and **530** are located to be biased to the fourth surface **514**.
(656) The third Halbach array **540** is disposed to face any one surface of the third and fourth surfaces **513** and **514** and the magnet unit **550** located adjacent to the any one surface. The space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the third Halbach array **540** and the any one surface.
(657) The third Halbach array **540** is located between the first surface **511** and the second surface **512**. In one embodiment, a third Halbach array **540** may be located at a central portion of the other surface of the third and fourth surfaces **513** and **514**.
(658) In other words, the shortest distance between the third Halbach array **540** and the first surface **511** and the shortest distance between the third Halbach array **540** and the second surface **512** may be the same.
(659) The third Halbach array **540** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and second Halbach arrays **520** and **530** and the magnet unit **550**. Since the process of enhancing the direction and magnetic

field of the magnetic field formed by the third Halbach array **540** is a well-known technique, a detailed description thereof will be omitted.
(660) In the illustrated embodiment, the third Halbach array **540** includes a first block **541**, a second block **542**, and a third block **543**. It will be understood that the plurality of magnetic materials constituting the third Halbach array **540** are named as the blocks **541**, **542**, and **543**, respectively.
(661) The first to third blocks **541**, **542**, and **543** may each be formed of a magnetic material. In one embodiment, the first to third blocks **541**, **542**, and **543** may each be provided as a permanent magnet, an electromagnet, or the like.
(662) The first to third blocks **541**, **542**, and **543** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **541**, **542**, and **543** are disposed in parallel in a direction in which the third surface **513** or the fourth surface **514** extends, that is, in the front-rear direction.
(663) The first block **541** is located at the most rear side. That is, the first block **541** is located adjacent to the first surface **511**. In addition, the third block **543** is located at the most front side. That is, the third block **543** is located adjacent to the second surface **512**. The second block **542** is located between the first block **541** and the third block **543**.
(664) In one embodiment, the blocks **541**, **542**, and **543** adjacent to each other may be in contact with each other.
(665) The second block **542** may be disposed to overlap the fixed contactors **22***a* and **22***b* in a direction toward the space part **515**, i.e., in the front-rear direction in the illustrated embodiment.
(666) Each of the blocks **541**, **542**, and **543** includes a plurality of surfaces.
(667) Specifically, the first block **541** includes the first inner surface **541***a* facing the second block **542** and the first outer surface **541***b* opposite to the second block **542**.
(668) The second block **542** includes the second inner surface **542***a* facing the space part **515** and the second outer surface **542***b* opposite to the space part **515**.
(669) The third block **543** includes the third inner surface **543***a* facing the second block **542** and the third outer surface **543***b* opposite to the second block **542**.
(670) The plurality of surfaces of each of the blocks **541**, **542**, and **543** may be magnetized according to a predetermined rule to configure a Halbach array.
(671) Specifically, the first to third inner surfaces **541***a*, **542***a*, and **543***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **541***b*, **542***b*, and **543***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **541***a*, **542***a*, and **543***a*.
(672) At this point, the first to third inner surfaces **541***a*, **542***a*, and **543***a* may be magnetized to the same polarity as the first to third inner surfaces **521***a*, **522***a*, and **523***a* of the first Halbach array **520** and the first to third inner surfaces **531***a*, **532***a*, and **533***a* of the second Halbach array **530**. In addition, the first to third inner surfaces **541***a*, **542***a*, and **543***a* may be magnetized to the same polarity as the opposing surface **552** of the magnet unit **550**.
(673) Similarly, the first to third outer surfaces **541***b*, **542***b*, and **543***b* may be magnetized to the same polarity as the first to third outer surfaces **521***b*, **522***b*, and **523***b* of the first Halbach array **520** and the first to third outer surfaces **531***b*, **532***b*, and **533***b* of the second Halbach array **530**. In addition, the first to third outer surfaces **541***b*, **542***b*, and **543***b* may be magnetized to the same polarity as the facing surface **551** of the magnet unit **550**.
(674) The magnet unit **550** forms a magnetic field by itself, or forms magnetic fields together with the first to third Halbach arrays **520**, **530**, and **540**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the magnet unit **550**.
(675) The magnet unit **550** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the magnet unit **550** may be provided as a permanent magnet, an electromagnet, or the like.
(676) The magnet unit **550** may be located adjacent to any one surface of the third and fourth

surfaces **513** and **514**. In one embodiment, the magnet unit **550** may be coupled to an inner side (i.e., the side in a direction toward the space part **515**) the any one surface.
(677) At this point, the magnet unit **550** is located on any one surface of the third and fourth surfaces **513** and **514**, to which the first and second Halbach arrays **520** and **530** are located to be biased.
(678) The magnet unit **550** is disposed to face the other surface of the third and fourth surfaces **513** and **514** with the space part **315** therebetween.
(679) In the embodiment illustrated in FIG. **25**, the magnet unit **550** is located adjacent to the third surface **513**. At this point, the first and second Halbach arrays **520** and **530** are located to be biased to the third surface **513**.
(680) In the embodiment illustrated in FIG. **26**, the magnet unit **550** is located adjacent to the fourth surface **514**. At this point, the first and second Halbach arrays **520** and **530** are located to be biased to the fourth surface **514**.
(681) The magnet unit **550** is disposed to face the other surface of the third and fourth surfaces **513** and **514** and the third Halbach array **540** located adjacent to the other surface. The space part **515**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **515** are located between the magnet unit **550** and the other surface.
(682) The magnet unit **550** is located between the first surface **511** and the second surface **512**. In one embodiment, the magnet unit **550** may be located at a central portion of any one surface of the third and fourth surfaces **513** and **514**.
(683) In other words, the shortest distance between the magnet unit **550** and the first surface **511** and the shortest distance between the magnet unit **550** and the second surface **512** may be the same.
(684) The magnet unit **550** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first to third Halbach arrays **520**, **530**, and **540**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first magnet unit **550** is well known in the art, a detailed description thereof will be omitted.
(685) The magnet unit **550** is formed to extend in one direction. In the illustrated embodiment, the magnet unit **550** is formed to extend in a direction in which the third surface **513** or the fourth surface **514** extends, that is, in the front-rear direction.
(686) The magnet unit **550** includes a plurality of surfaces.
(687) Specifically, the magnet unit **550** includes the facing surface **551** facing the space part **515** or the fixed contactor **22** and the opposing surface **552** opposite to the space part **515** or the fixed contactor **22**.
(688) Each surface of the magnet unit **550** may be magnetized according to a predetermined rule.
(689) Specifically, the facing surface **551** may be magnetized to the same polarity as the first to third outer surfaces **521***b*, **522***b*, and **523***b* of the first Halbach array **520** and the first to third outer surfaces **531***b*, **532***b*, and **533***b* of the second Halbach array **530**. In addition, the facing surface **551** may be magnetized to the same polarity as the first to third outer surfaces **541***b*, **542***b*, and **543***b* of the third Halbach array **540**.
(690) Similarly, the opposing surface **552** may be magnetized to the same polarity as the first to third inner surfaces **521***a*, **522***a*, and **523***a* of the first Halbach array **520** and the first to third inner surfaces **531***a*, **532***a*, and **533***a* of the second Halbach array **530**. In addition, the opposing surface **552** may be magnetized to the same polarity as the first to third inner surfaces **541***a*, **542***a*, and **543***a* of the third Halbach array **540**.
(691) At this point, it will be understood that the polarity of the facing surface **551** and the polarity of the opposing surface **552** are formed to be different from each other.
(692) Hereinafter, an arc path A.P formed by the arc path formation unit **500** according to the present embodiment will be described in detail with reference to FIGS. **27** and **28**.
(693) Referring to FIGS. **27** and **28**, the first to third inner surfaces **521***a*, **522***a*, and **523***a* of the first Halbach array **520** are magnetized to N poles. According to the above-described rule, the first

to third inner surfaces **531***a*, **532***a*, and **533***a* of the second Halbach array **530** and the first to third inner surfaces **541***a*, **542***a*, and **543***a* of the third Halbach array **540** are also magnetized to N poles.
(694) Furthermore, the facing surface **551** of the magnet unit **550** is magnetized to an S pole.
(695) Accordingly, magnetic fields that repel each other are formed between the first Halbach array **520** and the second Halbach array **530**. In addition, in the third Halbach array **540**, a magnetic field in a direction from the second inner surface **542***a* toward the facing surface **551** of the magnet unit **550** is formed.
(696) Accordingly, in the embodiment illustrated in FIG. **27**, the magnetic fields formed by the first to third Halbach arrays **520**, **530**, and **540** and the magnet unit **550** are formed toward the third surface **513**, i.e., toward the left side in the illustrated embodiment.
(697) In addition, in the embodiment illustrated in FIG. **28**, the magnetic fields formed by the first to third Halbach arrays **520**, **530**, and **540** and the magnet unit **550** are formed toward the fourth surface **514**, i.e., toward the right side in the illustrated embodiment.
(698) In the embodiment illustrated in FIG. **27**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(699) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(700) Accordingly, an arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(701) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(702) Accordingly, an arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(703) In the embodiment illustrated in FIG. **27**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(704) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(705) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(706) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(707) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(708) In the embodiment illustrated in FIG. **28**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(709) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(710) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(711) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(712) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(713) In the embodiment illustrated in FIG. **28**B, a direction of current is a direction from the first

fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(714) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(715) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(716) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(717) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(718) Although not illustrated in the drawing, when the polarity of each surface of the first to third Halbach arrays **520**, **530**, and **540** and the magnet unit **550** is changed, the direction of the magnetic field formed in each of first to third Halbach arrays **520**, **530**, and **540** and the magnet unit **550** is reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(719) That is, in the electric connection situation shown in FIG. **27**A, the electromagnetic force and the arc path A.P generated in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(720) Similarly, in the electric connection situation shown in FIG. **27**B, the electromagnetic force and the arc path A.P generated in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(721) In addition, in the electric connection situation shown in FIG. **28**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.
(722) Similarly, in the electric connection situation shown in FIG. **28**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.
(723) Accordingly, in the arc path formation unit **500** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the first to third Halbach arrays **520**, **530**, and **540** and the magnet unit **550** or the direction of the current flowing through the direct current relay **1**.
(724) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.
(6) Description of Arc Path Formation Unit **600** According to Yet Another Embodiment of Present Disclosure
(725) Hereinafter, an arc path formation unit **600** according to yet another embodiment of the present disclosure will be described in detail with reference to FIGS. **29** to **36**.
(726) Referring to FIGS. **29** to **32**, the arc path formation unit **600** according to the illustrated embodiment includes a magnet frame **610**, a first Halbach array **620**, a second Halbach array **630**, a first magnet unit **640**, and a second magnet unit **650**.
(727) The magnet frame **610** according to the present embodiment has the same structure and function as the magnet frame **110** according to the above-described embodiment. However, there is a difference in the arrangement method of the first Halbach array **620**, the second Halbach array **630**, the first magnet unit **640**, and the second magnet unit **650** disposed in the magnet frame **610**

according to the present embodiment.
(728) Accordingly, a description of the magnet frame **610** will be replaced with the description of the magnet frame **110** according to the above-described embodiment.
(729) In the illustrated embodiment, the plurality of magnetic materials constituting the first Halbach array **620** are continuously disposed side by side from the left side to the right side. That is, in the illustrated embodiment, the first Halbach array **620** is formed to extend in the left-right direction.
(730) The first Halbach array **620** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the first Halbach array **620** may form a magnetic field together with the second Halbach array **630**, the first and second magnet units **640** and **650**.
(731) The first Halbach array **620** may be located adjacent to any one surface of first and second surfaces **611** and **612**. In one embodiment, the first Halbach array **620** may be coupled to an inner side (i.e., the side in a direction toward a space part **615**) of the any one surface.
(732) In the embodiment illustrated in FIGS. **29** and **31**, the first Halbach array **620** is disposed on an inner side of the second surface **612** and adjacent to the second surface **612** and face the first magnet unit **640** located on an inner side of the first surface **611**.
(733) In the embodiment illustrated in FIGS. **30** and **32**, the first Halbach array **620** is disposed on the inner side of the first surface **611** and adjacent to the first surface **611** and faces the first magnet unit **640** located on the inner side of the second surface **612**.
(734) The space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the first Halbach array **620** and the second Halbach array **630**.
(735) In addition, the space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the first Halbach array **620** and the first magnet unit **640**.
(736) Furthermore, the space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the first Halbach array **620** and the second magnet unit **650**.
(737) The first Halbach array **620** may be located to be biased to any one surface of a third surface **613** and a fourth surface **614**. In the embodiment illustrated in FIGS. **29** and **30**, the first Halbach array **620** is located to be biased to the fourth surface **614**. In the embodiment illustrated in FIGS. **31** and **32**, the first Halbach array **620** is located to be biased to the third surface **613**.
(738) The any one surface, to which the first Halbach array **620** is located to be biased, may be a surface to which the second magnet unit **650** is located adjacent. In addition, the any one surface, to which the first Halbach array **620** is located to be biased, may be a surface opposite to a surface to which the second Halbach array **630** is located adjacent.
(739) The first Halbach array **620** may enhance the strength of the magnetic field formed by itself and the magnetic fields formed together with the second Halbach array **630** and the first and second magnet units **640** and **650**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first Halbach array **620** is a well-known technique, a detailed description thereof will be omitted.
(740) In the illustrated embodiment, the first Halbach array **620** includes a first block **621**, a second block **622**, and a third block **623**. It will be understood that the plurality of magnetic materials constituting the first Halbach array **620** are named as the blocks **621**, **622**, and **623**, respectively.
(741) The first to third blocks **621**, **622**, and **623** may each be formed of a magnetic material. In one embodiment, the first to third blocks **621**, **622**, and **623** may each be provided as a permanent magnet, an electromagnet, or the like.
(742) The first to third blocks **621**, **622**, and **623** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **621**, **622**, and **623** are disposed in parallel in a direction in which the first surface **611** extends, that is, in the left-right direction.

(743) The first block **621** is located adjacent to the any one surface of the third surface **613** and the fourth surface **614**. In addition, the third block **623** is located adjacent to the other surface of the third surface **613** and the fourth surface **614**. The second block **622** is located between the first block **621** and the third block **623**.
(744) In one embodiment, the blocks **621**, **622**, and **623** adjacent to each other may be in contact with each other.
(745) The second block **622** may be disposed to overlap the first magnet unit **640** and any one of the fixed contactors **22***a* and **22***b* in a direction toward the first magnet unit **640** or the space part **615**, i.e., in the front-rear direction in the illustrated embodiment.
(746) Each of the blocks **621**, **622**, and **623** includes a plurality of surfaces.
(747) Specifically, the first block **621** includes a first inner surface **621***a* facing the second block **622** and a first outer surface **621***b* opposite to the second block **622**.
(748) The second block **622** includes a second inner surface **622***a* facing the space part **615** or the first magnet unit **640** and a second outer surface **622***b* opposite to the space part **615** or the first magnet unit **640**.
(749) The third block **623** includes a third inner surface **623***a* facing the second block **622** and a third outer surface **623***b* opposite to the second block **622**.
(750) The plurality of surfaces of each of the blocks **621**, **622**, and **623** may be magnetized according to a predetermined rule to configure a Halbach array.
(751) Specifically, the first to third inner surfaces **621***a*, **622***a*, and **623***a* may be magnetized to the same polarity. Similarly, the first to third outer surfaces **621***b*, **622***b*, and **623***b* are magnetized to a polarity different from the polarity of the first to third inner surfaces **621***a*, **622***a*, and **623***a*.
(752) At this point, the first to third inner surfaces **621***a*, **622***a*, and **623***a* may be magnetized to the same polarity as first to third inner surfaces **631***a*, **632***a*, and **633***a* of the second Halbach array **630** and a first facing surface **641** of the first magnet unit **640**. In addition, the first to third inner surfaces **621***a*, **622***a*, and **623***a* may be magnetized to the same polarity as a second opposing surface **652** of the second magnet unit **650**.
(753) Similarly, the first to third outer surfaces **621***b*, **622***b*, and **623***b* may be magnetized to the same polarity as first to third outer surfaces **631***b*, **632***b*, and **633***b* of the second Halbach array **630** and a first opposing surface **642** of the first magnet unit **640**. In addition, the first to third outer surfaces **621***b*, **622***b*, and **623***b* may be magnetized to the same polarity as a second facing surface **651** of the second magnet unit **650**.
(754) In the illustrated embodiment, the plurality of magnetic materials constituting the second Halbach array **630** are continuously disposed side by side from the front side to the rear side. That is, in the illustrated embodiment, the second Halbach array **630** is formed to extend in the front-rear direction.
(755) The second Halbach array **630** may form a magnetic field together with another magnetic material. In the illustrated embodiment, the second Halbach array **630** may form magnetic fields together with the first Halbach array **620** and the first and second magnet units **640** and **650**.
(756) The second Halbach array **630** may be located adjacent to the other surface of the third and fourth surfaces **613** and **614**. In one embodiment, the second Halbach array **630** may be coupled to an inner side (i.e., the side in a direction toward the space part **615**) of the other surface.
(757) The second Halbach array **630** may be located adjacent to a surface opposite to any one surface to which the first Halbach array **620** and the first magnet unit **640** are located to be biased.
(758) In the embodiment illustrated in FIGS. **29** and **30**, the second Halbach array **630** is disposed on an inner side of the fourth surface **614** and adjacent to the fourth surface **614** and faces the second magnet unit **650** located on an inner side of the third surface **613**.
(759) In the embodiment illustrated in FIGS. **31** and **32**, the second Halbach array **630** is disposed on the inner side of the third surface **613** and adjacent to the third surface **613** and faces the second magnet unit **650** located on the inner side of the fourth surface **614**.

(760) The space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the second Halbach array **630** and the first Halbach array **620**.
(761) In addition, the space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the second Halbach array **630** and the first magnet unit **640**.
(762) Furthermore, the space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the second Halbach array **630** and the second magnet unit **650**.
(763) The second Halbach array **630** is located between the first surface **611** and the second surface **612**. In one embodiment, the second Halbach array **630** may be located at a central portion of any one surface of the third and fourth surfaces **613** and **614**.
(764) In other words, the shortest distance between the second Halbach array **630** and the first surface **611** and the shortest distance between the second Halbach array **630** and the second surface **612** may be the same.
(765) The second Halbach array **630** may enhance the strength of the magnetic field formed by itself and the magnetic fields formed together with the first Halbach array **620** and the first and second magnet units **640** and **650**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second Halbach array **630** is a well-known technique, a detailed description thereof will be omitted.
(766) In the illustrated embodiment, the second Halbach array **630** includes a first block **631**, a second block **632**, and a third block **633**. It will be understood that the plurality of magnetic bodies constituting the second Halbach array **630** are named as the blocks **631**, **632**, and **633**, respectively.
(767) The first to third blocks **631**, **632**, and **633** may each be formed of a magnetic material. In one embodiment, the first to third blocks **631**, **632**, and **633** may each be provided as a permanent magnet, an electromagnet, or the like.
(768) The first to third blocks **631**, **632**, and **633** may be disposed in parallel in one direction. In the illustrated embodiment, the first to third blocks **631**, **632**, and **633** are disposed in parallel in a direction in which the third surface **613** or the fourth surface **614** extends, that is, in the front-rear direction.
(769) The first block **631** is located at the most front side. That is, the first block **631** is located adjacent to the first surface **611**. In addition, the third block **633** is located at the most rear side. That is, the third block **633** is located adjacent to the second surface **612**. The second block **632** is located between the first block **631** and the third block **633**.
(770) In one embodiment, the blocks **631**, **632**, and **633** adjacent to each other may be in contact with each other.
(771) The second block **632** may be disposed to overlap the fixed contactors **22*a*** and **22*b*** in a direction toward the second Halbach array **630** or the space part **615**, i.e., in the front-rear direction in the illustrated embodiment.
(772) Each of the blocks **631**, **632**, and **633** includes a plurality of surfaces.
(773) Specifically, the first block **631** includes the first inner surface **631*a*** facing the second block **632** and the first outer surface **631*b*** opposite to the second block **632**.
(774) The second block **632** includes the second inner surface **632*a*** facing the space part **615** or the second magnet unit **650** and the second outer surface **632*b*** opposite to the space part **615** or the second magnet unit **650**.
(775) The third block **633** includes the third inner surface **633*a*** facing the second block **632** and the third outer surface **633*b*** opposite to the second block **632**.
(776) The plurality of surfaces of each of the blocks **631**, **632**, and **633** may be magnetized according to a predetermined rule to configure a Halbach array.
(777) Specifically, the first to third inner surfaces **631*a***, **632*a***, and **633*a*** may be magnetized to the

same polarity. Similarly, the first to third outer surfaces **631***b*, **632***b*, and **633***b* may be magnetized to a polarity different from the polarity of the first to third inner surfaces **631***a*, **632***a*, and **633***a*.
(778) At this point, the first to third inner surfaces **631***a*, **632***a*, and **633***a* may be magnetized to the same polarity as the first to third inner surfaces **621***a*, **622***a*, and **623***a* of the first Halbach array **620** and the first facing surface **641** of the first magnet unit **640**. In addition, the first to third inner surfaces **631***a*, **632***a*, and **633***a* may be magnetized to the same polarity as the second opposing surface **652** of the second magnet unit **650**.
(779) Similarly, the first to third outer surfaces **631***b*, **632***b*, and **633***b* may be magnetized to the same polarity as the first to third outer surfaces **621***b*, **622***b*, and **623***b* of the first Halbach array **620** and the first opposing surface **642** of the first magnet unit **640**. In addition, the first to third outer surfaces **631***b*, **632***b*, and **633***b* may be magnetized to the same polarity as the second facing surface **651** of the second magnet unit **650**.
(780) The first magnet unit **640** forms a magnetic field by itself, or forms magnetic fields together with the first and second Halbach arrays **620** and **630** and the second magnet unit **650**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the first magnet unit **640**.
(781) The first magnet unit **640** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the first magnet unit **640** may be provided as a permanent magnet, an electromagnet, or the like.
(782) The first magnet unit **640** may be located adjacent to the other surface of the first and second surfaces **611** and **612**. In one embodiment, the first magnet unit **640** may be coupled to an inner side (i.e., the side in a direction toward the space part **615**) of the other surface.
(783) At this point, the first magnet unit **640** is located to be biased to any one surface of the third and fourth surfaces **613** and **614**, to which the second magnet unit **650** is located adjacent. In other words, the first magnet unit **640** is located to be biased to a surface opposite to the other surface of the third and fourth surfaces **613** and **614**, to which the second Halbach array **630** is located adjacent.
(784) The first magnet unit **640** is disposed to face the first Halbach array **620** with the space part **615** therebetween.
(785) In the embodiment illustrated in FIGS. **29** and **31**, the first magnet unit **640** is located adjacent to the first surface **611**. In addition, in the embodiment illustrated in FIGS. **30** and **32**, the first magnet unit **640** is located adjacent to the second surface **612**.
(786) In the embodiment illustrated in FIGS. **29** and **30**, the first magnet unit **640** is located to be biased to the third surface **613**. In the embodiment illustrated in FIGS. **31** and **32**, the first magnet unit **640** is located to be biased to the fourth surface **614**.
(787) The first magnet unit **640** is disposed to face the other surface of the first and second surfaces **611** and **612** and the first Halbach array **620** located adjacent to the other surface. The space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the first magnet unit **640** and the other surface.
(788) In one embodiment, the first magnet unit **640** may be disposed to overlap the first Halbach array **620** and any one of the fixed contactors **22***a* and **22***b* in the front-rear direction.
(789) The first magnet unit **640** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and second Halbach arrays **620** and **630** and the second magnet unit **650**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the first magnet unit **640** is well known in the art, a detailed description thereof will be omitted.
(790) The first magnet unit **640** is formed to extend in one direction. In the illustrated embodiment, the first magnet unit **640** is formed to extend in a direction in which the first surface **611** or the second surface **612** extends, that is, in the left-right direction.
(791) The first magnet unit **640** includes a plurality of surfaces.

(792) Specifically, the first magnet unit **640** includes the first facing surface **641** facing the space part **615** or the fixed contactor **22** and the first opposing surface **642** opposite to the space part **615** or the fixed contactor **22**.
(793) Each surface of the first magnet unit **640** may be magnetized according to a predetermined rule.
(794) Specifically, the first facing surface **641** may be magnetized to the same polarity as the first to third inner surfaces **621***a*, **622***a*, and **623***a* of the first Halbach array **620** and the first to third inner surfaces **631***a*, **632***a*, and **633***a* of the second Halbach array **630**. In addition, the first facing surface **641** may be magnetized to the same polarity as the second opposing surface **652** of the second magnet unit **650**.
(795) Similarly, the first opposing surface **642** may be magnetized to the same polarity as the first to third outer surfaces **621***b*, **622***b*, and **623***b* of the first Halbach array **620** and the first to third outer surfaces **631***b*, **632***b*, and **633***b* of the second Halbach array **630**. In addition, the first opposing surface **642** may be magnetized to the same polarity as the second facing surface **651** of the second magnet unit **650**.
(796) At this point, it will be understood that the polarity of the first facing surface **641** and the polarity of the first opposing surface **642** are formed to be different from each other.
(797) The second magnet unit **650** forms a magnetic field by itself, or forms magnetic fields together with the first and second Halbach arrays **620** and **630** and the first magnet unit **640**. An arc path A.P may be formed inside the arc chamber **21** by the magnetic field formed by the second magnet unit **650**.
(798) The second magnet unit **650** may be provided in any form capable of being magnetized to form a magnetic field. In one embodiment, the second magnet unit **650** may be provided as a permanent magnet, an electromagnet, or the like.
(799) The second magnet unit **650** may be located adjacent to any one surface of the third and fourth surfaces **613** and **614**. In one embodiment, the second magnet unit **650** may be coupled to an inner side (i.e., the side in a direction toward the space part **615**) of the any one surface.
(800) At this point, the second magnet unit **650** is located on any one surface of the third and fourth surfaces **613** and **614**, to which the first Halbach array **620** and the first magnet unit **640** are located to be biased.
(801) The second magnet unit **650** is disposed to face the other surface of the third and fourth surfaces **613** and **614** and the second Halbach array **630**, which is located adjacent to the other surface, with the space part **615** therebetween.
(802) In the embodiment illustrated in FIGS. **29** and **30**, the second magnet unit **650** is located adjacent to the third surface **613**. At this point, the first Halbach array **620** and the first magnet unit **640** are located to be biased to the third surface **613**. In addition, the second Halbach array **630** is located adjacent to the fourth surface **614**.
(803) In the embodiment illustrated in FIGS. **31** and **32**, the second magnet unit **650** is located adjacent to the fourth surface **614**. At this point, the first Halbach array **620** and the first magnet unit **640** are located to be biased to the fourth surface **614**. In addition, the second Halbach array **630** is located to be biased to the third surface **613**.
(804) The second magnet unit **650** is disposed to face the other surface of the third and fourth surfaces **613** and **614** and the second Halbach array **630** located adjacent to the other surface. The space part **615**, and the fixed contactor **22** and the movable contactor **43** accommodated in the space part **615** are located between the second magnet unit **650** and the other surface.
(805) The second magnet unit **650** is located between the first surface **611** and the second surface **612**. In one embodiment, the second magnet unit **650** may be located at a central portion of any one surface of the third and fourth surfaces **613** and **614**.
(806) In other words, the shortest distance between the second magnet unit **650** and the first surface **611** and the shortest distance between the second magnet unit **650** and the second surface **612** may

be the same.
(807) The second magnet unit **650** can enhance the strength of the magnetic field formed by itself and the strength of the magnetic fields formed together with the first and second Halbach arrays **620** and **630** and the first magnet unit **640**. Since the process of enhancing the direction and magnetic field of the magnetic field formed by the second magnet unit **650** is well known in the art, a detailed description thereof will be omitted.
(808) The second magnet unit **650** is formed to extend in one direction. In the illustrated embodiment, the second magnet unit **650** is formed to extend in a direction in which the third surface **613** or the fourth surface **614** extends, that is, in the front-rear direction.
(809) The second magnet unit **650** includes a plurality of surfaces.
(810) Specifically, the second magnet unit **650** includes the second facing surface **651** facing the space part **615** or the fixed contactor **22** and the second opposing surface **652** opposite to the space part **615** or the fixed contactor **22**.
(811) Each surface of the second magnet unit **650** may be magnetized according to a predetermined rule.
(812) Specifically, the second facing surface **651** may be magnetized to the same polarity as the first to third outer surfaces **621***b*, **622***b*, and **623***b* of the first Halbach array **620** and the first to third outer surfaces **631***b*, **632***b*, and **633***b* of the second Halbach array **630**. In addition, the second facing surface **651** may be magnetized to the same polarity as the first opposing surface **642** of the first magnet unit **640**.
(813) Similarly, the second opposing surface **652** may be magnetized to the same polarity as the first to third inner surfaces **621***a*, **622***a*, and **623***a* of the first Halbach array **620** and the first to third inner surfaces **631***a*, **632***a*, and **633***a* of the second Halbach array **630**. In addition, the second opposing surface **652** may be magnetized to the same polarity as the first facing surface **641** of the first magnet unit **640**.
(814) At this point, it will be understood that the polarity of the second facing surface **651** and the polarity of the second opposing surface **652** are formed to be different from each other.
(815) Hereinafter, an arc path A.P formed by the arc path formation unit **600** according to the present embodiment will be described in detail with reference to FIGS. **33** to **36**.
(816) Referring to FIGS. **33** to **36**, the first to third inner surfaces **621***a*, **622***a*, and **623***a* of the first Halbach array **620** are magnetized to N poles. According to the above-described rule, the first to third inner surfaces **631***a*, **632***a*, and **633***a* of the second Halbach array **630** and the first facing surface **641** of the first magnet unit **640** are also magnetized to N poles. At this point, the second facing surface **651** of the second magnet unit **650** is magnetized to an S pole.
(817) Accordingly, magnetic fields that repel each other are formed between the first Halbach array **620** and the first magnet unit **640**. In addition, a magnetic field in a direction from the second inner surface **632***a* toward the second facing surface **651** is formed between the second Halbach array **630** and the second magnet unit **650**.
(818) Accordingly, in the embodiment illustrated in FIGS. **33** and **34**, the magnetic fields formed by the first and second Halbach arrays **620** and **630** and the first and second magnet units **640** and **650** are formed toward the third surface **613**, i.e., toward the left side in the illustrated embodiment.
(819) In addition, in the embodiment illustrated in FIGS. **35** and **36**, the magnetic fields formed by the first and second Halbach arrays **620** and **630** and the first and second magnet units **640** and **650** are formed toward the fourth surface **614**, i.e., toward the right side in the illustrated embodiment.
(820) In the embodiment illustrated in FIGS. **33**A and **34**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(821) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(822) Accordingly, an arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also

formed toward the front left side.
(823) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(824) Accordingly, an arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(825) In the embodiment illustrated in FIGS. **33**B and **34**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(826) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(827) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(828) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(829) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(830) In the embodiment illustrated in FIGS. **35**A and **36**A, a direction of current is a direction from the second fixed contactor **22***b* to the first fixed contactor **22***a* via the movable contactor **43**.
(831) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the rear left side.
(832) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the rear left side.
(833) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the front right side.
(834) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the front right side.
(835) In the embodiment illustrated in FIGS. **35**B and **36**B, a direction of current is a direction from the first fixed contactor **22***a* to the second fixed contactor **22***b* via the movable contactor **43**.
(836) When the Fleming's left-hand rule is applied to the first fixed contactor **22***a*, an electromagnetic force generated in the vicinity of the first fixed contactor **22***a* is formed toward the front left side.
(837) Accordingly, the arc path A.P generated in the vicinity of the first fixed contactor **22***a* is also formed toward the front left side.
(838) Similarly, when the Fleming's left-hand rule is applied to the second fixed contactor **22***b*, an electromagnetic force generated in the vicinity of the second fixed contactor **22***b* is formed toward the rear right side.
(839) Accordingly, the arc path A.P generated in the vicinity of the second fixed contactor **22***b* is also formed toward the rear right side.
(840) Although not illustrated in the drawing, when the polarity of each surface of the first and second Halbach arrays **620** and **630** and the first and second magnet units **640** and **650** is changed, the directions of the magnetic fields formed in the first and second Halbach arrays **620** and **630** and the first and second magnet units **640** and **650** are reversed. Accordingly, the generated electromagnetic force and the arc path A.P are also formed so that the front-rear direction thereof is reversed.
(841) That is, in the electric connection situation shown in FIGS. **33**A and **34**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear

left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.

(842) Similarly, in the electric connection situation shown in FIGS. **33**B and **34**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.

(843) In addition, in the electric connection situation shown in FIGS. **35**A and **36**A, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the front left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the rear right side.

(844) Similarly, in the electric connection situation shown in FIGS. **35**B and **36**B, the electromagnetic force and the arc path A.P in the vicinity of the first fixed contactor **22***a* are formed toward the rear left side. In addition, the electromagnetic force and the arc path A.P generated in the vicinity of the second fixed contactor **22***b* are formed toward the front right side.

(845) Accordingly, in the arc path formation unit **600** according to the present embodiment, the electromagnetic force and the arc path A.P may be formed in a direction away from the central part C regardless of the polarity of each of the first and second Halbach arrays **620** and **630** and the first and second magnet units **640** and **650** or the direction of the current flowing through the direct current relay **1**.

(846) Accordingly, damage to each component of the direct current relay **1** disposed adjacent to the central part C can be prevented. Furthermore, since the generated arc can be quickly discharged to the outside, operational reliability of the direct current relay **1** can be improved.

(847) Although it has been described above with reference to preferred embodiments of the present disclosure, it will be understood that those skilled in the art are able to variously modify and change the present disclosure without departing from the spirit and scope of the disclosure described in the claims below. **1**: direct current relay **10**: frame part **11**: upper frame **12**: lower frame **13**: insulating plate **14**: supporting plate **20**: opening/closing part **21**: arc chamber **22**: fixed contactor **22***a*: first fixed contactor **22***b*: second fixed contactor **23**: sealing member **30**: core part **31**: fixed core **32**: movable core **33**: yoke **34**: bobbin **35**: coil **36**: return spring **37**: cylinder **40**: movable contactor part **41**: housing **42**: cover **43**: movable contactor **44**: shaft **45**: elastic part **100**: arc path formation unit according to one embodiment of present disclosure **110**: magnet frame **111**: first surface **112**: second surface **113**: third surface **114**: fourth surface **115**: space part **120**: first Halbach array **121**: first block **121***a*: first inner surface **121***b*: first outer surface **122**: second block **122***a*: second inner surface **122***b*: second outer surface **123**: third block **123***a*: third inner surface **123***b*: third outer surface **130**: second Halbach array **131**: first block **131***a*: first inner surface **131***b*: first outer surface **132**: second block **132***a*: second inner surface **132***b*: second outer surface **133**: third block **133***a*: third inner surface **133***b*: third outer surface **140**: third Halbach array **141**: first block **141***a*: first inner surface **141***b*: first outer surface **142**: second block **142***a*: second inner surface **142***b*: second outer surface **143**: Third block **143***a*: third inner surface **143***b*: third outer surface **200**: arc path formation unit according to another embodiment of present disclosure **210**: magnet frame **211**: first surface **212**: second surface **213**: third surface **214**: fourth surface **215**: space part **220**: first Halbach array **221**: first block **221***a*: first inner surface **221***b*: first outer surface **222**: second block **222***a*: second inner surface **222***b*: second outer surface **223**: third block **223***a*: third inner surface **223***b*: third outer surface **230**: second Halbach array **231**: first block **231***a*: first inner surface **231***b*: first outer surface **232**: second block **232***a*: second inner surface **232***b*: second outer surface **233**: third block **233***a*: third inner surface **233***b*: third outer surface **240**: magnet unit **241**: facing surface **242**: opposing surface **300**: arc path formation unit according still another embodiment of present disclosure **310**: magnet frame **311**: first surface **312**: second surface **313**: third surface **314**: fourth surface **315**: space part **320**: first Halbach array **321**: first block **321***a*: first inner surface **321***b*: first outer surface **322**: second block **322***a*: second inner surface **322***b*: second

outer surface **323**: third block **323***a*: third inner surface **323***b*: third outer surface **330**: first magnet unit **331**: first facing surface **332**: first opposing surface **340**: second magnet unit **341**: second facing surface **342**: second opposing surface **400**: arc path formation unit according yet another embodiment of present disclosure **410**: magnet frame **411**: first surface **412**: second surface **413**: third surface **414**: fourth surface **415**: space part **420**: first Halbach array **421**: first block **421***a*: first inner surface **421***b*: first outer surface **422**: second block **422***a*: second inner surface **422***b*: second outer surface **423**: Third block **423***a*: third inner surface **423***b*: third outer surface **430**: first magnet unit **431**: first facing surface **432**: first opposing surface **440**: second magnet unit **441**: second facing surface **442**: second opposing surface **500**: arc path formation unit according to yet another embodiment of present disclosure **510**: magnet frame **511**: first surface **512**: second surface **513**: third surface **514**: fourth surface **515**: space part **520**: first Halbach array **521**: first block **521***a*: first inner surface **521***b*: first outer surface **522**: second block **522***a*: second inner surface **522***b*: second outer surface **523**: Third block **523***a*: third inner surface **523***b*: third outer surface **530**: second Halbach array **531**: first block **531***a*: first inner surface **531***b*: first outer surface **532**: second block **532***a*: second inner surface **532***b*: second outer surface **533**: Third block **533***a*: third inner surface **533***b*: third outer surface **540**: third Halbach array **541**: first block **541***a*: first inner surface **541***b*: first outer surface **542**: second block **542***a*: second inner surface **542***b*: second outer surface **543**: third block **343***a*: third inner surface **543***b*: third outer surface **550**: magnet unit **551**: facing surface **552**: opposing surface **600**: arc path formation unit according yet another embodiment of present disclosure **610**: magnet frame **611**: first surface **612**: second surface **613**: third surface **614**: fourth surface **615**: space part **620**: first Halbach array **621**: first block **621***a*: first inner surface **621***b*: first outer surface **622**: second block **622***a*: second inner surface **622***b*: second outer surface **623**: Third block **623***a*: third inner surface **623***b*: third outer surface **630**: second Halbach array **631**: first block **631***a*: first inner surface **631***b*: first outer surface **632**: second block **632***a*: second inner surface **632***b*: second outer surface **633**: Third block **633***a*: third inner surface **640**: first magnet unit **641**: first facing surface **642**: first opposing surface **650**: second magnet unit **651**: second facing surface **652**: second opposing surface **1000**: direct current relay according to related art **1100**: fixed contact according to related art **1200**: movable contact according to related art **1300**: permanent magnet according to related art **1310**: first permanent magnet according to related art **1320**: second permanent magnet according to related art C: central part of each of space parts **115**, **215**, **315**, **415**, **515**, and **615** A.P: arc path

Claims

1. An arc path formation unit comprising: a magnet frame having a space part, in which a fixed contactor and a movable contactor are accommodated, formed therein; and a Halbach array located in the space part of the magnet frame and configured to form a magnetic field in the space part, wherein a length of the space part in a first direction is formed to be greater than a length thereof in a second direction, wherein the magnet frame includes: a first surface and a second surface which extend in the first direction, are disposed to face each other, and are configured to surround a portion of the space part; and a third surface and a fourth surface which extend in the second direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, wherein the fixed contactor includes: a first fixed contactor located closer to the fourth surface than to the third surface; and a second fixed contactor located closer to the third surface than to the fourth surface; and wherein the Halbach array includes: a first Halbach array located adjacent to the first surface, located closer to the fourth surface than to the third surface; a second Halbach array located adjacent to the second surface, located closer to the fourth surface than to the third surface, and disposed to face the first Halbach array with the first fixed contactor therebetween; and a third Halbach array located adjacent to the third surface, wherein each of the first Halbach array, the

second Halbach array, and the third Halbach array comprises: a first block; a second block disposed adjacent to the first block, wherein with a first magnetic pole of the first block is directed toward the second block, wherein the first magnetic pole of the second block is directed toward the first fixed contactor; and a third block disposed adjacent to the second block, with the first magnetic pole of the third block directed toward the second block, wherein: the first Halbach array is located on a first half of the first surface that is located closer to the fourth surface than to the third surface; a second half of the first surface extends from the first half of the first surface to the third surface and is free of magnets; the second Halbach array located on a first half of the second surface, located closer to the fourth surface than to the third surface; a second half of the second surface extends from the first half of the second surface to the third surface and is free of magnets; the first fixed contactor is disposed between the first half of the first surface and the first half of the second surface and not disposed between the second half of the first surface and the second half of the second surface; and the second fixed contactor is disposed between the second half of the first surface and the second half of the second surface and not disposed between the first half of the first surface and the first half of the second surface.

2. The arc path formation unit of claim 1, comprising a magnet unit provided separately from the Halbach array, located in the space part of the magnet frame to form a magnetic field in the space part, located adjacent to the fourth surface, and disposed to overlap the first fixed contactor and the third Halbach array in the first direction.

3. A direct current relay comprising: a plurality of fixed contactors located to be spaced apart from each other in a first direction; a movable contactor configured to be brought into contact with or separated from the fixed contactors; a magnet frame having a space part, in which the fixed contactors and the movable contactor are accommodated, formed therein; and a Halbach array located in the space part of the magnet frame and configured to form a magnetic field in the space part, wherein a length of the space part in the first direction is formed to be greater than a length thereof in a second direction, wherein the magnet frame includes: a first surface and a second surface which extend in the first direction, are disposed to face each other, and are configured to surround a portion of the space part; and a third surface and a fourth surface which extend in the second direction, are continuous with the first surface and the second surface, respectively, are disposed to face each other, and are configured to surround a remaining portion of the space part, wherein the plurality of fixed contactors includes: a first fixed contactor located closer to the fourth surface than to the third surface; and a second fixed contactor located closer the third surface than to the fourth surface, and wherein the Halbach array includes: a first Halbach array located adjacent to the first surface, located closer to the fourth surface than to the third surface; a second Halbach array located adjacent to the second surface, located closer to the fourth surface than to the third surface, and disposed to face the first Halbach array with the first fixed contactor therebetween; and a third Halbach array located adjacent to the third surface, wherein each of the first Halbach array, the second Halbach array, and the third Halbach array includes: a first block; a second block disposed adjacent to the first block, wherein with a first magnetic pole of the first block is directed toward the second block, wherein the first magnetic pole of the second block is directed toward the first fixed contactor; and a third block disposed adjacent to the second block, with the first magnetic pole of the third block directed toward the second block, wherein: the first Halbach array is located on a first half of the first surface that is located closer to the fourth surface than to the third surface; a second half of the first surface extends from the first half of the first surface to the third surface and is free of magnets; the second Halbach array located on a first half of the second surface, located closer to the fourth surface than to the third surface; a second half of the second surface extends from the first half of the second surface to the third surface and is free of magnets; the first fixed contactor is disposed between the first half of the first surface and the first half of the second surface and not disposed between the second half of the first surface and the second half of the second surface; and the second fixed contactor is disposed between the second half of the first

surface and the second half of the second surface and not disposed between the first half of the first surface and the first half of the second surface.

4. The direct current relay of claim 3, comprising a magnet unit provided separately from the Halbach array, located in the space part of the magnet frame to form a magnetic field in the space part, located adjacent to the third surface, and disposed to overlap the first fixed contactor and the third Halbach array in the first direction, wherein: among surfaces of the magnet unit, a surface facing the space part has a second magnetic polarity, the second magnetic polarity being opposite to the first magnetic polarity.